

SeoJin Jeong — Cycloidal Quasi-Direct Drive Actuator (및 관련 액추에이터 글)

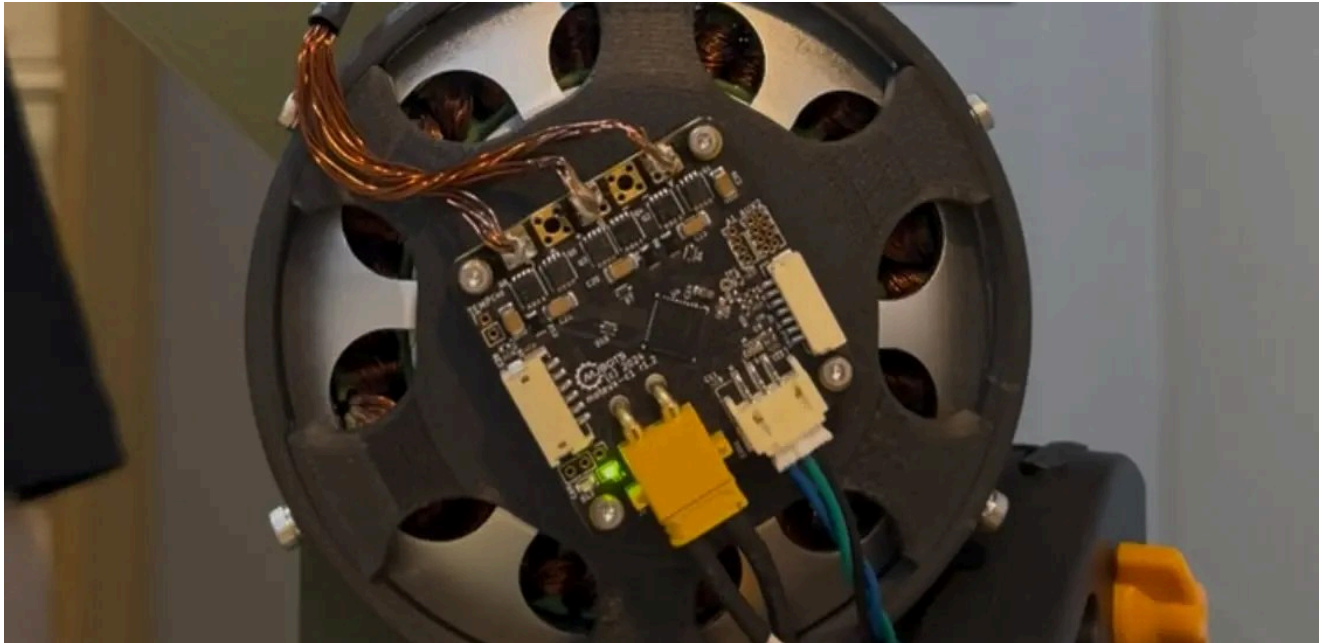
웹페이지 글·그림 발췌 — 2026-09-05 수집

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A Cycloidal Quasi-Direct Drive Actuator Design, Fabrication and Control

출처: <https://jeongseojin.github.io/blog/2025/qdd-actuator/>



Actuator image 2

▶ 임베드: <https://www.youtube.com/embed/m4KiFf9MHiw>

Background: Motivation and Evolution

This Project builds upon my previous project project, ‘A Low-Cost 3D-Printed Proprioceptive Actuator for HRI’, and represents a significant step forward in performance and design.

In the ealier actuators(v1, v2 and v3), these exhibited insufficient torque density, which limited their application in dynamic and agile robotic systems. The primary cause was the use of small size off-the-shelf BLDC drone motors; their small air-gap radius and small torque-density could not generate the necessary torque.

To address this problem, I initiated this project, focusing on maximizing torque density, compactness and system efficiacy. By adopting a large-diameter “pancake” BLDC motor to maximize the air-gap radius for high torque density and integrating a internal cycloidal reducer for compactness, I enhanced the actuator’s dynamics and agility. Inspired by the MIT Mini Cheetah’s design philosophy, this project demonstrates a transition toward a more robust and high-performance Quasi-Direct Drive (QDD) system.

1. Abstract

Rapid advancements in dynamic humanoid and quadruped robots have surged recently. However, high-performance actuators remain largely proprietary, restricting access for academic researchers and open-source communities. This project aims to democratize dynamic robotics by developing an open-source Quasi-Direct Drive (QDD) actuator tailored for dynamic robots

Traditional high-ratio gearboxes suffer from poor back-drivability, low responsiveness and limited transmission transparency, making them unsuitable for safe Human-Robot Interaction (HRI) and uneven terrain locomotion. To address this, I propose a custom-designed QDD actuator featuring a 10:1 cycloidal reducer. This low-reduction design minimizes mechanical impedance, enabling back-drivable, high responsiveness and proprioceptive torque sensing via FOC algorithms without additional sensors.

The current prototype utilizes a 3D-printed and aluminium structure to ensure low cost and accessibility. And the performance of my actuator; Maximum Torque: 8.8Nm, Maximum Velocity: 22rad/s and Nominal Velocity 12.6rad/s under 24V. I want this project contributes to the robotic community by providing a scalable, affordable, and the actuation solution.



Actuator image 1



Actuator image 2

2. Actuator Design: Why QDD Actuator?

2.1 The Limitations of Traditional Actuators

Conventional robotic actuators typically utilize high gear reduction ratios (e.g., 1:100 or higher) to amplify torque. However, this approach introduces significant drawbacks for dynamic robots.

A high gear ratio leads to high mechanical impedance, primarily due to increased friction and reflected inertia compared to low-ratio systems [1].

This high mechanical impedance makes the motor response sluggish and susceptible to damage from sudden external impacts, as the actuators are too stiff to react compliantly. In other words, the actuator exhibits poor back-drivability and low responsiveness [2]. Consequently, these properties hinder the robot's ability to interact safely with the environment, particularly in Human-Robot Interaction (HRI) scenarios or during dynamic locomotion.

As robots become more integrated into our daily lives, physical interaction and cooperation will become increasingly common. In these scenarios, traditional actuators pose a significant safety risk. Since robots with stiff actuators struggle to sense external forces [3], they can unintentionally injure people due to their lack of compliance. Furthermore, regarding locomotion, such robots cannot flexibly adapt to unpredictable environments, such as rough or uneven terrain, leading to instability.

Additionally, the complex dynamics & mechanical impedance of such actuators make them difficult to model accurately in physical simulators, creating a significant Sim-to-Real gap. Such modeling inaccuracies often lead to the failure of hardware deployment, particularly when using policies trained via Reinforcement Learning [4].

2.2 The QDD System Solution

To address these issues, I adopted a Quasi-Direct Drive (QDD) architecture. A QDD system typically features a low gear reduction ratio, ranging from 3:1 to 10:1. By positioning itself between Direct Drive (1:1) and traditional high-ratio drives (1:50 or higher), it combines the structural advantages of both systems.

The low gear ratio significantly minimizes friction and reflected inertia, making the actuator inherently compliant. This compliance ensures the system is smoothly back-drivable and highly responsive to external interactions, effectively mitigating the stiffness drawbacks of conventional actuators.

Consequently, this improved back-drivability and responsiveness enable the robot to safely interact with humans and flexibly adapt to uneven terrain during dynamic locomotion.

At this point, a critical question arises: “How can the actuator generate sufficient torque for dynamic motions with such a low gear ratio?” The following section details the design strategy used to resolve this trade-off.

2.3 Motor Design Optimization

To achieve high torque density suitable for a Quasi-Direct Drive (QDD) system, I focused on the geometric parameters of the motor. According to the motor design principles outlined in the MIT Cheetah research, the continuous torque generation capability () is approximated by the following equation [1], [5]:

$$\tau \propto \sigma \cdot l \cdot s \cdot t \cdot r \cdot g^2$$

Where:

- : Shear stress density (magnetic flux density
×
current loading)
- : Stack length of the motor
- : Air gap radius

As shown in the equation, torque is linearly proportional to the stack length () but proportional to the square of the air gap radius (g^2)

). This implies that increasing the motor diameter is significantly more efficient for boosting torque than increasing its length. Based on this principle, I selected a stator with a large diameter (8110 size) to maximize the air gap radius, thereby securing sufficient torque even with a low gear reduction ratio. Furthermore, this large-diameter stator design provides a sufficient hollow center which can be utilized to embed the reducer gear, improving the compactness of the actuator and torque density.

2.4 Transmission Selection: Why Cycloidal Reducer?

In dynamic legged locomotion, the actuator must withstand high impact loads caused by ground reaction forces. While Harmonic Drives are industry standards for zero-backlash precision, their flexible spline mechanism is fragile under shock loads. And structurally, it's not suitable for low-ratio reducer.

Similarly, Planetary Gearboxes, though common, exhibit inherent backlash. It works well with aluminum gears, but becomes significantly fragile when 3D printed. In a 3D-printed planetary system,

the stress concentrates on individual small gear teeth, making them prone to catastrophic failure under sudden external forces.

To ensure robustness, I selected a Cycloidal Reducer architecture. According to Sensinger's research, this mechanism distributes the load across multiple lobes simultaneously. This load-sharing capability provides significantly higher Shock Resistance compared to involute gears or fragile harmonic drives, making it the ideal candidate for a fully 3D-printed transmission [6], [7], [8].

3. System Architecture & Components

3.1 Mechanical Design (Cycloidal Reducer)

A custom Cycloidal Quasi-Direct Drive Actuator is designed to ensure high torque density, compactness, and compliance.

I utilized Onshape 3D CAD to design a dual-disc cycloidal mechanism. The two cycloidal discs are arranged with a

180

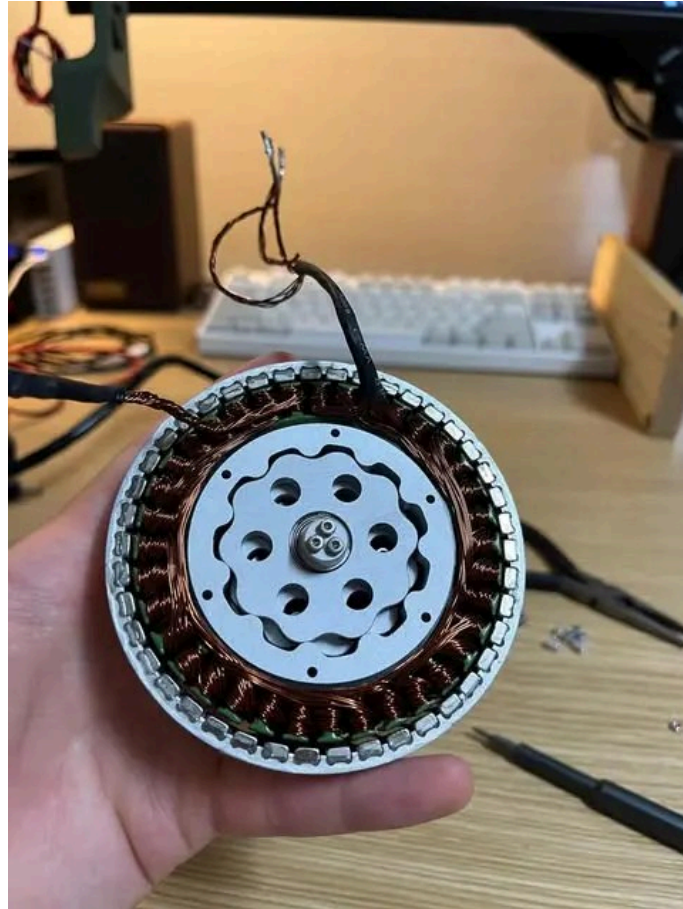
°
phase offset. This configuration effectively cancels out the radial forces and vibrations induced by the eccentric input shaft, ensuring smooth operation.

To maximize efficiency, I integrated rollers into the output pins. Unlike simple sliding contacts, these rollers minimize internal friction at the output stage, contributing to the system's back-drivability.



Actuator image 2

For the current prototype, the gears, shafts and rotor are manufactured from CNC-machined Aluminum to verify the design with a high strength-to-weight ratio. (Note: The final goal of this project is to optimize the design for a fully 3D-printed, low-cost actuator for dynamic robots.)



Actuator image 1



Actuator image 2

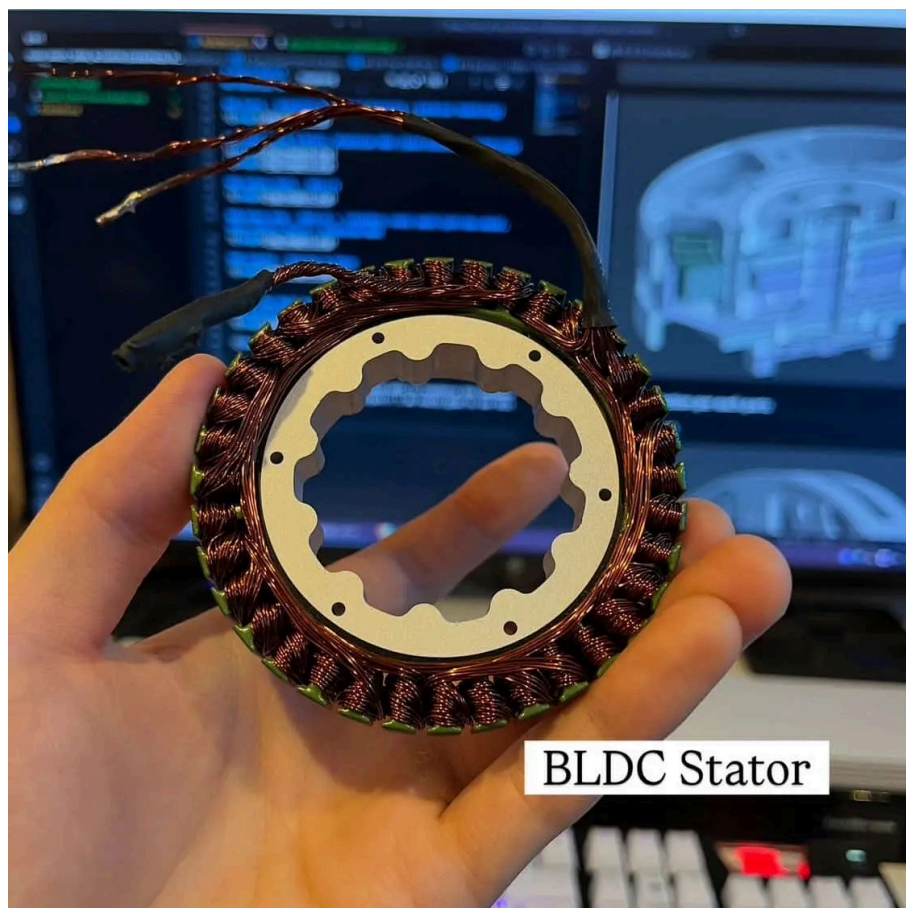
3.2 Electromagnetic Design (Custom BLDC)

To optimize torque density within the compact housing, I integrated a custom-built frameless Brushless DC (BLDC) motor instead of using a standard motor.

For the stator, I utilized a standard 8110 stator core. To achieve the desired current capacity and fill factor, the stator was hand-wound using 0.4mm enameled copper wire. I applied a Wye (Star) termination with 6 parallel strands and 5 turns per tooth, following the optimal winding scheme calculated via open-source tools [9].



Actuator image 1



BLDC Stator

Actuator image 2

The motor adopts a 36N42P configuration (36 slots, 42 poles) to maximize torque output. For the rotor, 42 N52-grade Neodymium magnets were installed. These magnets were precisely bonded using high-strength epoxy (JB Weld) in an alternating polarity pattern (N-S-N-S) to maximize magnetic flux density and ensure structural integrity under high rotation speeds. Crucially, the rotor geometry was optimized to achieve a minimal air gap of 0.5mm. This tight clearance maximizes the magnetic flux linkage between the rotor and stator, thereby significantly enhancing the electromagnetic force and overall torque efficiency.



Actuator image 1

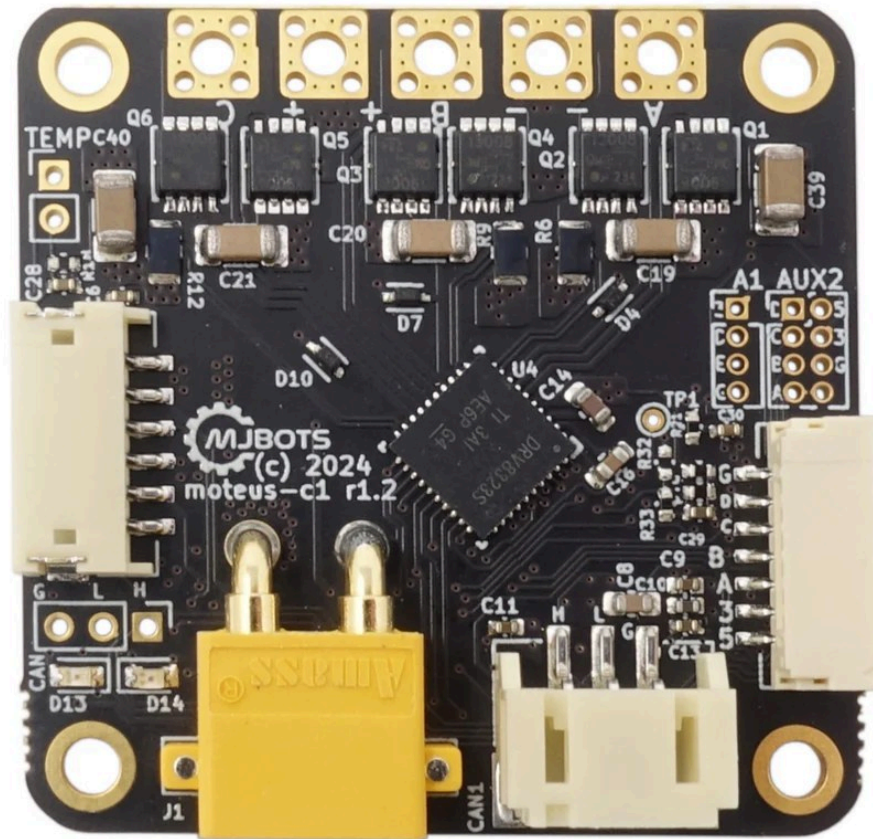


Actuator image 2

3.3 Electronics & Sensors

For precise torque control and dynamic response, I integrated the Moteus-c1 controller (mjbots). This controller is widely adopted in the dynamic robotics community for its compact form factor and high bandwidth.

It implements Field Oriented Control (FOC), which is essential for smooth torque generation and the proprioceptive capabilities mentioned earlier. With a wide input range (10-51V) and 20A peak phase current, it provides sufficient power capacity to drive the custom-wound 8110 stator to its full potential.



Actuator image 2

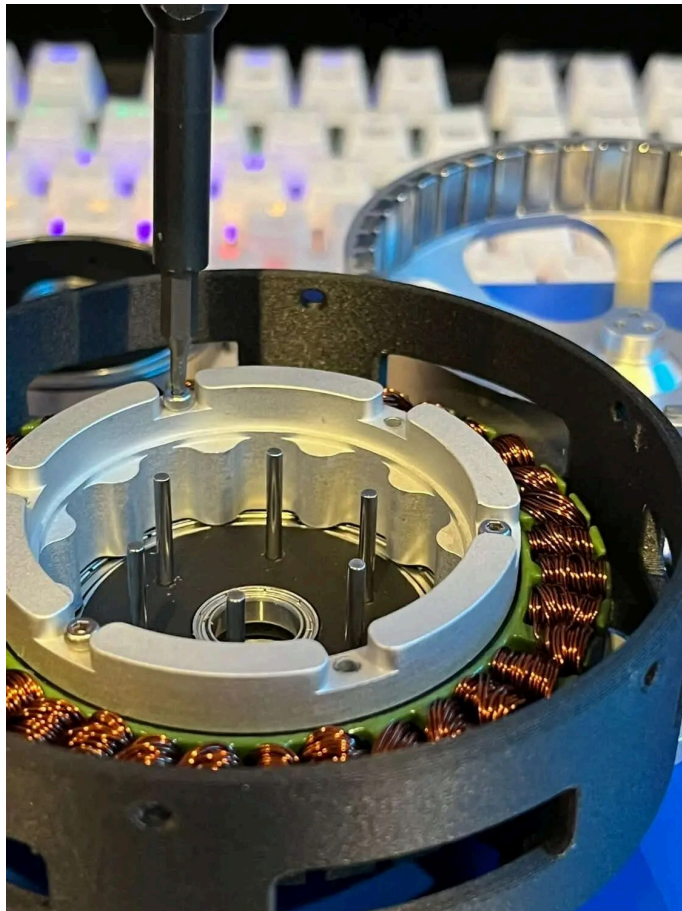
4. Assembly

Precise assembly is critical to minimize backlash and ensure the longevity of the reducer. I implemented distinct tolerance strategies based on the material properties of each component.

4.1 Bearing Installation & Fits

For the housing components printed in PA-CF12 (Carbon Fiber Nylon), I designed a 0.1mm interference fit. This accounts for the material's slight compliance and thermal shrinkage during printing, ensuring a secure press-fit without cracking the part.

Conversely, for the CNC-machined Aluminum parts, a tighter 0.02mm interference fit was applied due to the metal's rigidity. To prevent any micro-movements or slippage under high torque loads, I reinforced these metal-to-bearing interfaces with a thin application of JB Weld epoxy.



Actuator image 1



Actuator image 2

4.2 Output Mechanism

The output transmission relies on smooth rolling contact. I installed six M2x20mm stainless steel shafts to serve as the carrier pins. These shafts were precision-aligned to allow the external rollers to rotate freely, minimizing friction at the output stage while maintaining structural rigidity.

5. Control & Validation

To validate the actuator's performance, I conduct a torque estimation experiment using load-cell. The torque was applied from 1 Nm up to 10 Nm, while performing maximum torque of 8.8Nm. Maximum speed is measured 22rad/s and Nominal speed is 12.6rad/s



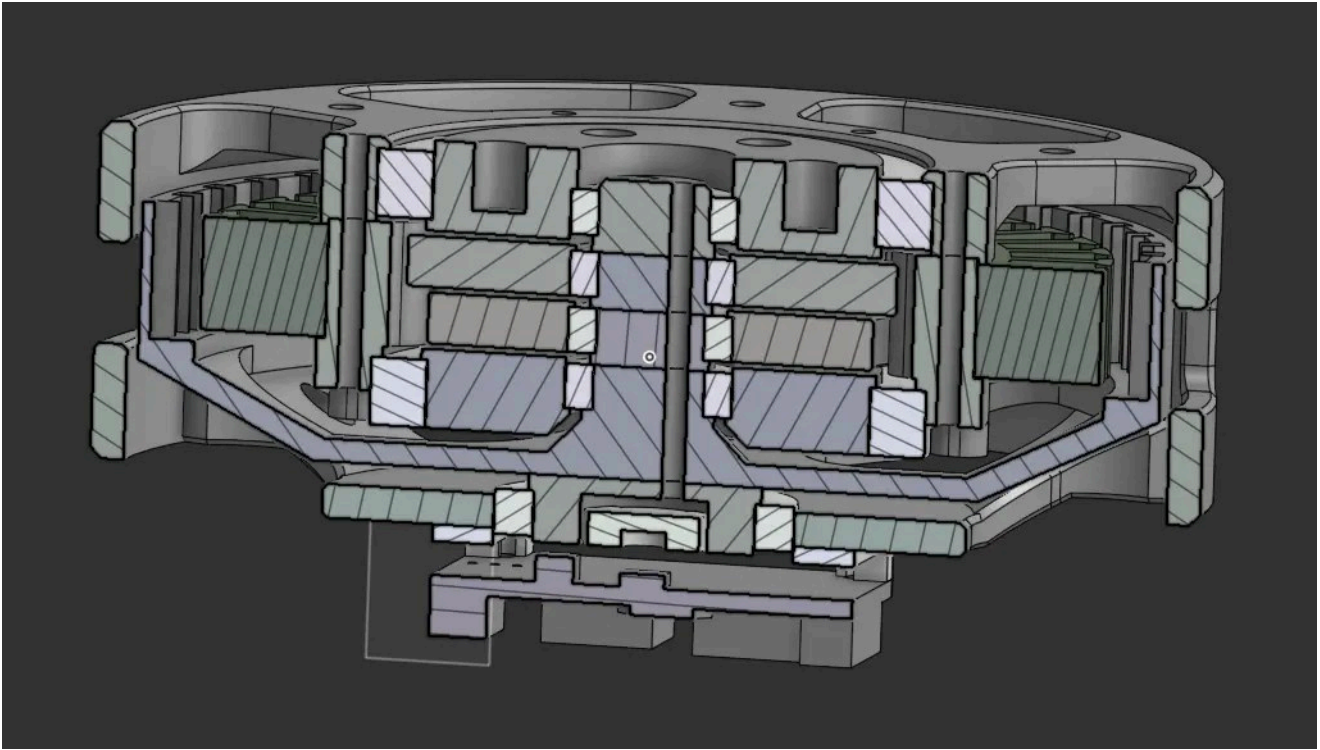
Actuator image 2

6. Limitations & Future Works

6.1 Limitations of the Current Prototype

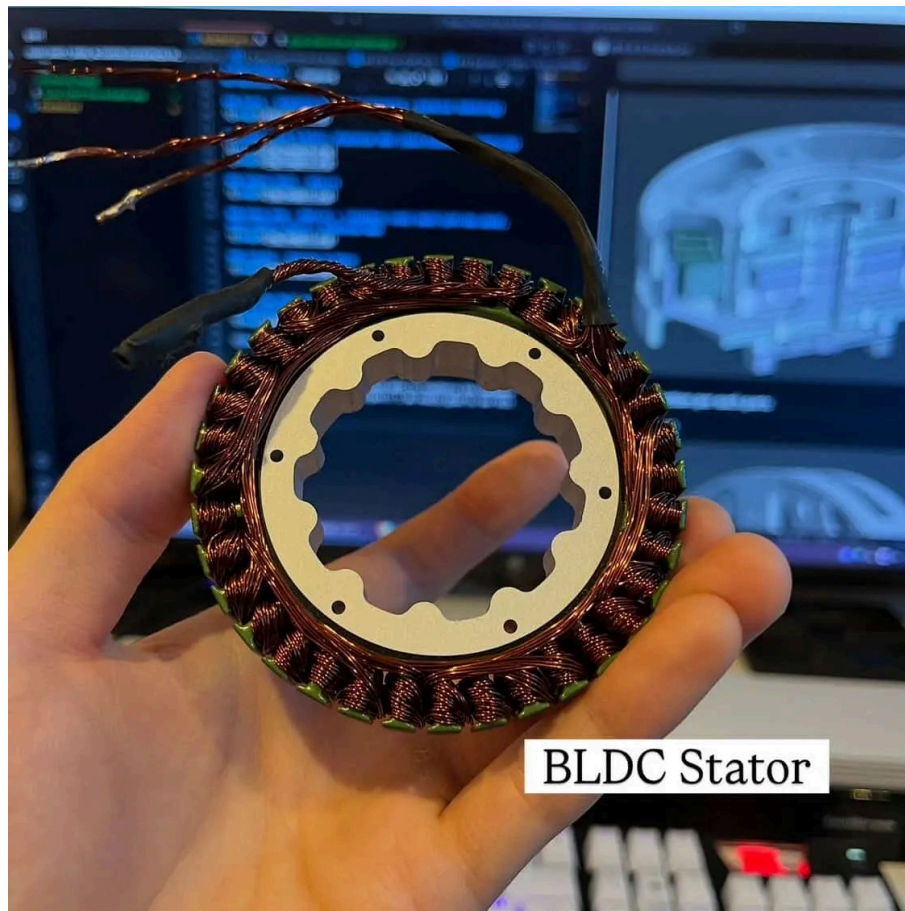
Although this first prototype works successfully, four primary limitations were identified regarding the actuator's performance and structural dynamics through the testing and assembly process.

- Rotor Shaft Rigidity(misalignment issue): > The current aluminium rotor shaft is not a single component, this is constructed as a multi-part assembly fastened with bolts and nuts, rather than a single monolithic part. Consequently, the strong electromagnetic forces between the rotor and stator occasionally cause the central shaft to misalign during rotation, leading to an uneven air gap.



Actuator image 2

- Absence of a Rotor Back-Iron: > The current rotor design does not incorporate a ferromagnetic back-iron or a Halbach array arrangement. I initially overlooked the critical role of the back-iron which concentrates the magnetic flux inward to maximize magnetic force. Without it, the actuator suffers from flux leakage, resulting in a maximum torque which is insufficient for highly dynamic robotic applications.
- Low Copper Fill Factor: > Since I hand-wound the stator, the copper wires are relatively sparsely packed compared to tightly wound commercial motors. This lower slot fill factor significantly limits the current-carrying capacity and overall torque density.



Actuator image 2

- Structural Limits for Modularity: >The design of the housing is not suitable for 3d-printed modular actuator. Because the housings are 3D printed, I have to reinforce and modify the design to withstand external mechanical disturbances.

6.2 Future Works

Although the first prototype functions adequately, I don't think this prototype is suitable for dynamic robotic systems because of the issues that I mentioned above. Therefore I'm planning to modify the actuator design and improve torque density to achieve successful dynamic robotic actuator in the next iteration (QDD Actuator Ver2)

Moving forward, I aim to integrate these custom QDD actuators into a mid-size bipedal robot and bi-manipulator to validate their performance in a complete system. Those robotic systems strongly require the compliant and dynamic behavior of the QDD Actuator. This project will serve as a crucial step in verifying the scalability of my hardware design while providing a physical testbed for implementing advanced locomotion control algorithms and manipulation tasks.

7. Credits

- Design, Engineering and Documentation: Seo Jin Jeong
- Manufacturing Sponsor: JLCCNC
- Motor Controller: Moteus-c1 Controller (mjbots)
- Youtube : Engineering SeoJin
- Instagram : engineering.seojin_n.n

8. References

- [1] S. Seok et al., “Design Principles for Energy-Efficient Legged Locomotion and Implementation on the MIT Cheetah Robot,” in *IEEE/ASME Transactions on Mechatronics*, vol. 20, no. 3, pp. 1117-1129, June 2015, doi: 10.1109/TMECH.2014.2339013.
- [2] T. Zhu, J. Hooks and D. Hong, “Design, Modeling, and Analysis of a Liquid Cooled Proprioceptive Actuator for Legged Robots,” in *2019 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, Hong Kong, China, 2019, pp. 36-43, doi: 10.1109/AIM.2019.8868596.
- [3] P. M. Wensing, A. Wang, S. Seok, D. Otten, J. Lang and S. Kim, “Proprioceptive Actuator Design in the MIT Cheetah: Impact Mitigation and High-Bandwidth Physical Interaction for Dynamic Legged Robots,” in *IEEE Transactions on Robotics*, vol. 33, no. 3, pp. 509-522, June 2017, doi: 10.1109/TRO.2016.2640183.
- [4] J. Tan et al., “Sim-to-Real: Learning Agile Locomotion for Quadruped Robots,” in *Proc. Robotics: Science and Systems (RSS)*, 2018, arXiv:1804.10332.
- [5] MIT Biomimetics Robotics Lab, “Proprioceptive Actuator Design,” biomimetics.mit.edu. [Online]. Available: <https://biomimetics.mit.edu/research/1a9ba04b-d200-4743-b8fc-bf231f3231f0>. [Accessed: Jan. 23, 2026].
- [6] J. W. Sensinger, “Unified Approach to Cycloid Drive Profile, Stress, and Efficiency Optimization,” *ASME Journal of Mechanical Design*, vol. 132, no. 2, p. 024503, Feb. 2010, doi: 10.1115/1.4000832.
- [7] K. Lee, S. Hong, and J. H. Oh, “Development of a Lightweight and High-efficiency Compact Cycloidal Reducer for Legged Robots,” *International Journal of Precision Engineering and Manufacturing*, vol. 21, no. 1, pp. 63-71, Jan. 2020, doi: 10.1007/s12541-019-00215-9.
- [8] J. W. Sensinger and J. H. Lipsey, “Cycloid vs. Harmonic Drives for Use in High Ratio, Single Stage Robotic Transmissions,” in *2012 IEEE International Conference on Robotics and Automation (ICRA)*, Saint Paul, MN, USA, 2012, pp. 4130-4135, doi: 10.1109/ICRA.2012.6224739.
- [9] “Homebuilt Electric Motors - Calculator,” bavaria-direct.co.za. [Online]. Available: <https://www.bavaria-direct.co.za/scheme/calculator/>. [Accessed: Jan. 23, 2026].

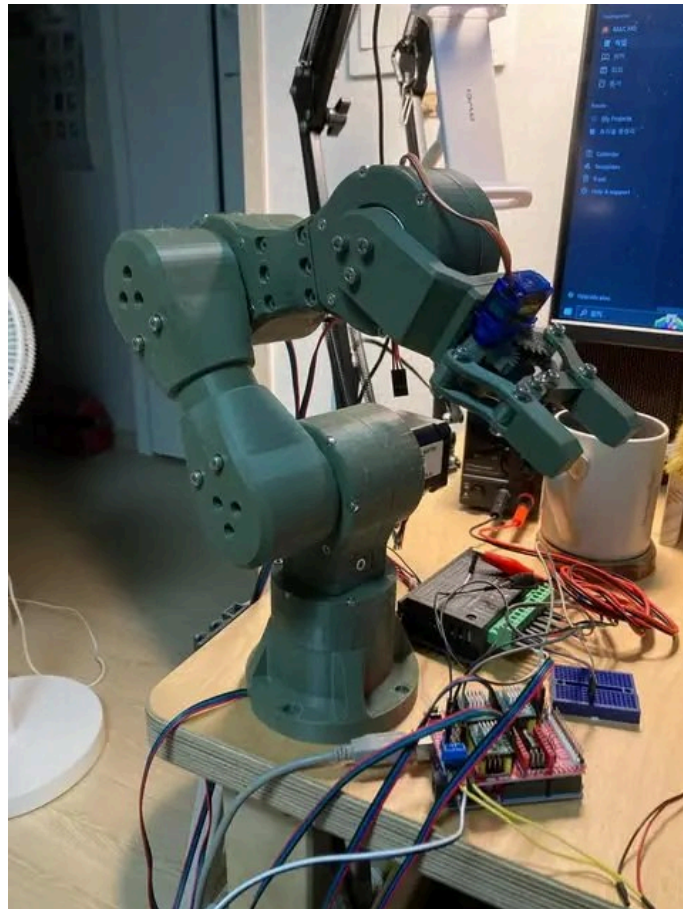
A Low-Cost Modular 3D-Printed Proprioceptive Actuator for HRI

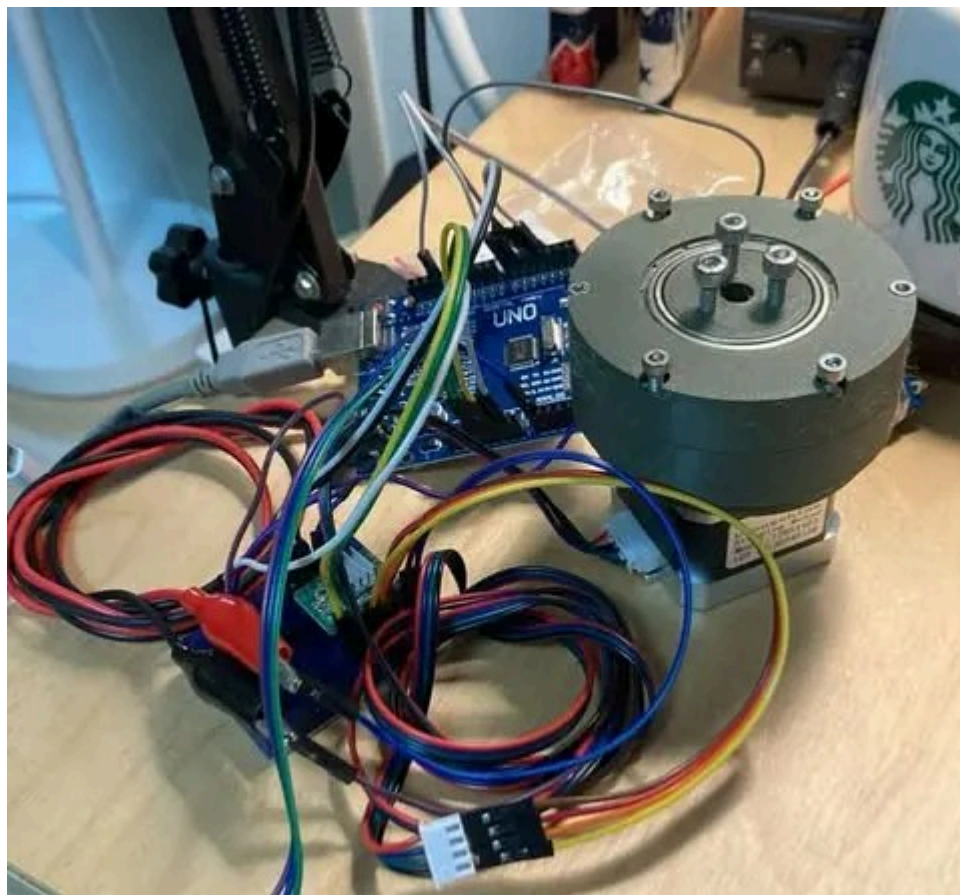
출처: <https://jeongseojin.github.io/blog/2025/modular-3d-printed-proprioceptive-actuators/>

view this page in github with resources

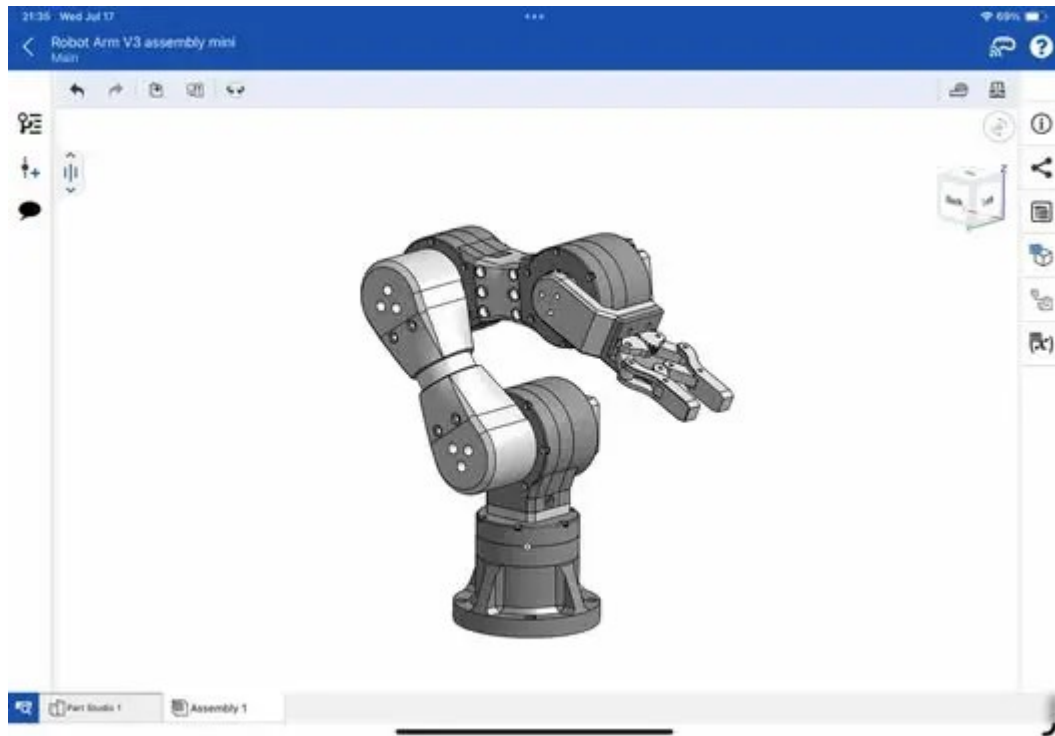
Background: Motivation and Evolution

In my previous project, 'Modular Cycloidal Actuator-based 4DoF Cooperative Robotic Arm', featuring 27:1 gear ratio, fully 3D-printed cycloidal reducer and precise position control using nama17 stepper motor with a4988 motor driver. However, the lack of back-drivability of the actuators and torque-based control make my robotic arm significantly stiff, restricting Human-Robot Interaction(HRI).









To address this issue, I developed 'A Low-Cost 3D-Printed Proprioceptive Actuator', utilizing bldc drone motors, FOC controller for precise torque/position control and proprioceptive sensing of current of motor. And I designed Quasi-Direct Drive(QDD) reducer(low gear ratio reducer) and Impedance Control demonstrating high dynamics, agility and compliance properties for HRI.

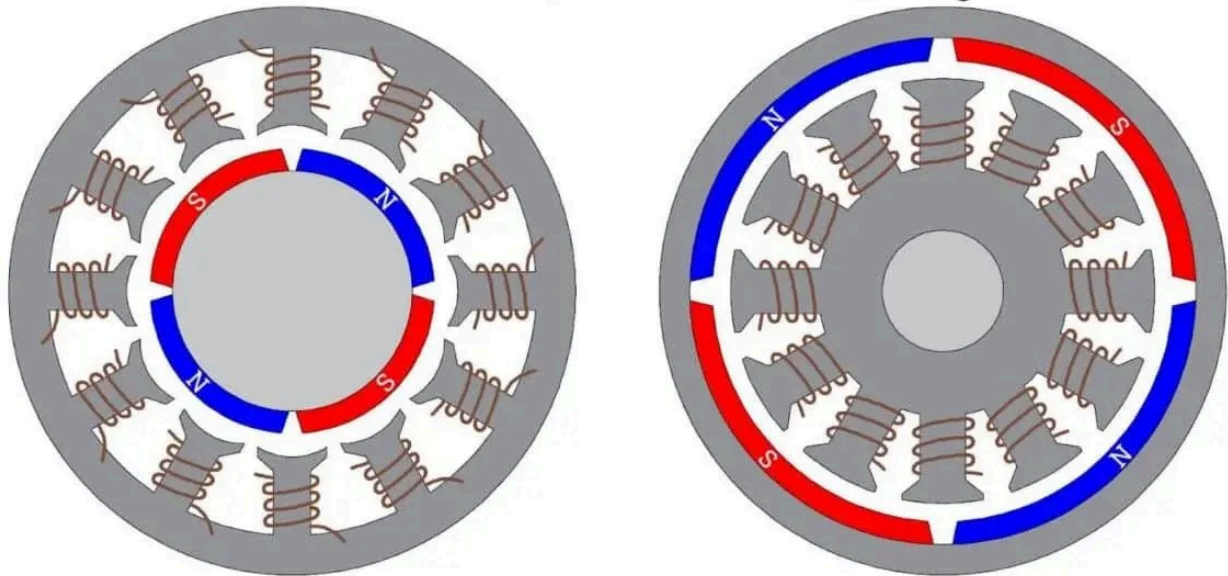
BLDC Motor

BLDC Motor Brushless DC (BLDC) motors are highly suitable for dynamic and agile robotic actuators due to their exceptional torque density, high velocity and acceleration, back-drivability, and proprioceptive sensing capabilities.

A BLDC Motor is composed of two main parts: the stator and the rotor. the stator produce a rotating electromagnetic field via a motor controller(ESC), and the rotor(permanent magnet) follows this field to produce rotation without any brushes like standard DC Motor.

This brushless design eliminates mechanical friction from brushes, leading to higher efficiency, longer lifespan and most importantly, the ability to estimate torque through current sensing—a key feature for proprioceptive control in legged robots.

Inrunner and Outrunner BLDC Motor Diagrams

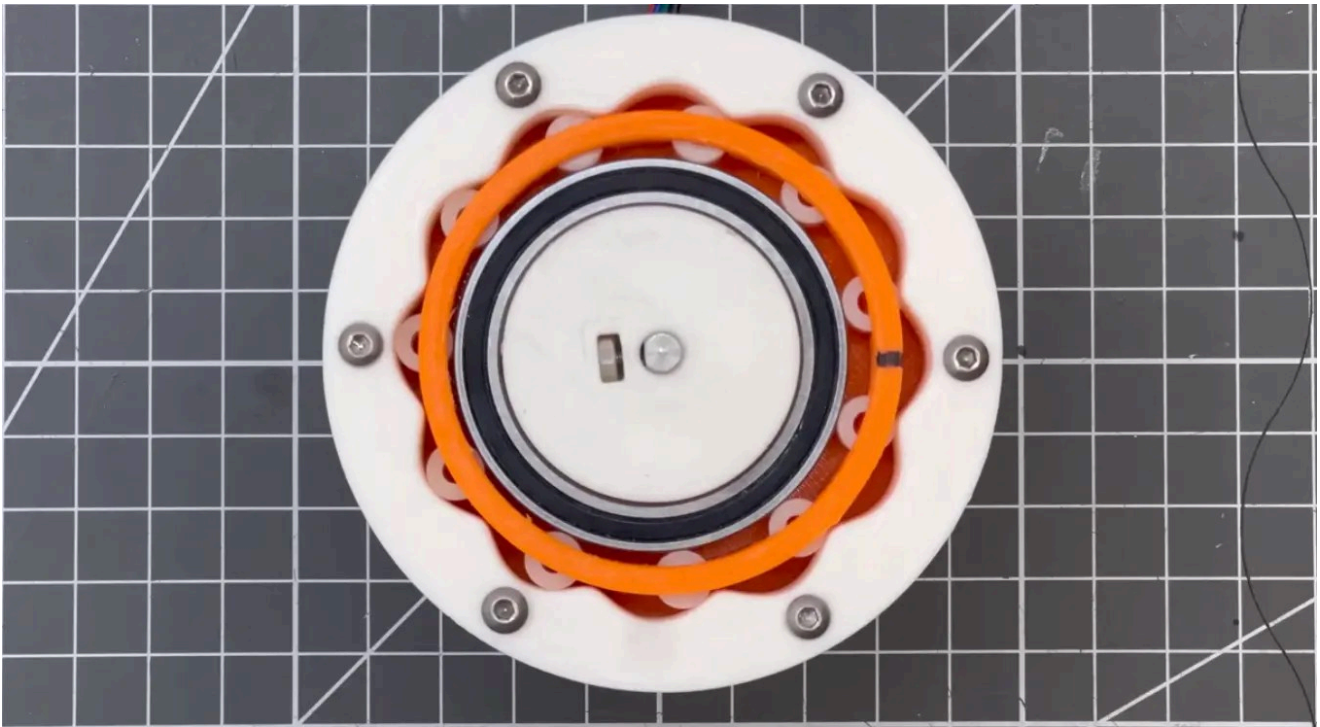


Source: Etonm Motor

I optend for an Outrunner BLDC Motor because a larger air-gap radius allows the motor to produce higher torque. Since the air-gap radius is directly correlates with torque density. Thus this design is ideal for high-performance actuators. For more technical detials, please refer to my other post: Paper Review: 'Design Principles for Energy Efficient Legged Locomotion and Implementation on the MIT Cheetah Robot'."

Reducer

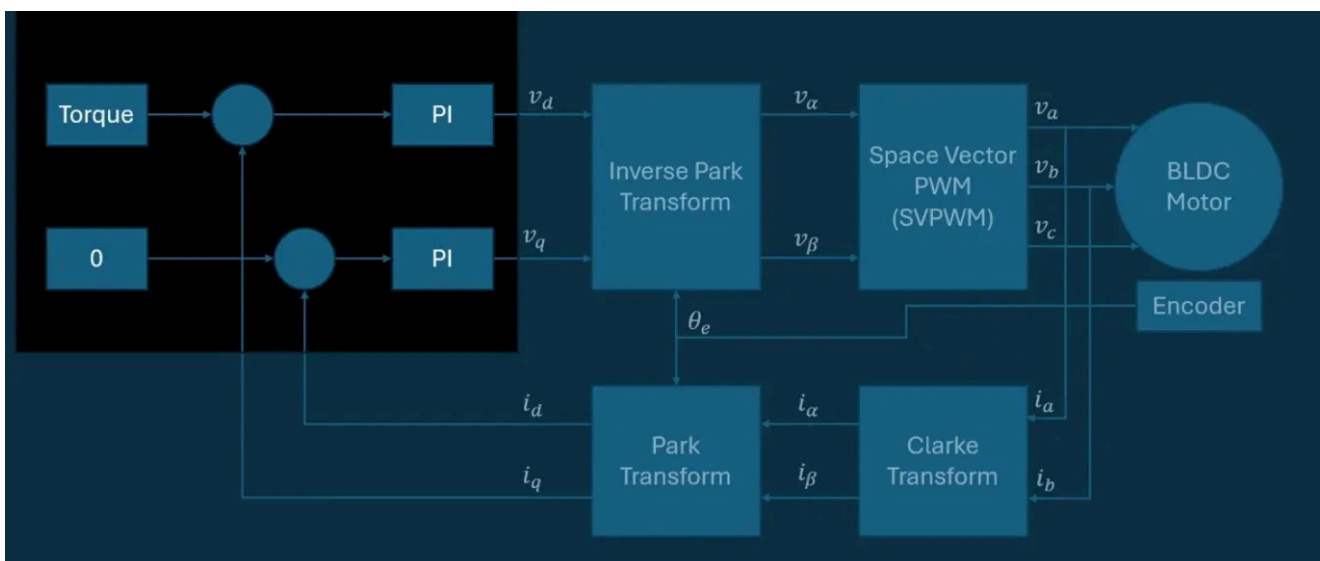
I was inspired by Mishin Machine-Wave Drive with Rolling Elements which is cycloidal-like reducer using ball-elements sharing as an open-source in Onshape. It's compact, simplfied with a few components and robust.



Source: Mishin Machine

FOC Controller

I used FOC Controller, 'drv8302 motor driver' to control the BLDC Motor. Unlike standard Electric Speed Controller(ESC), FOC Controller enables position/torque control etc.



Source: SIROJU

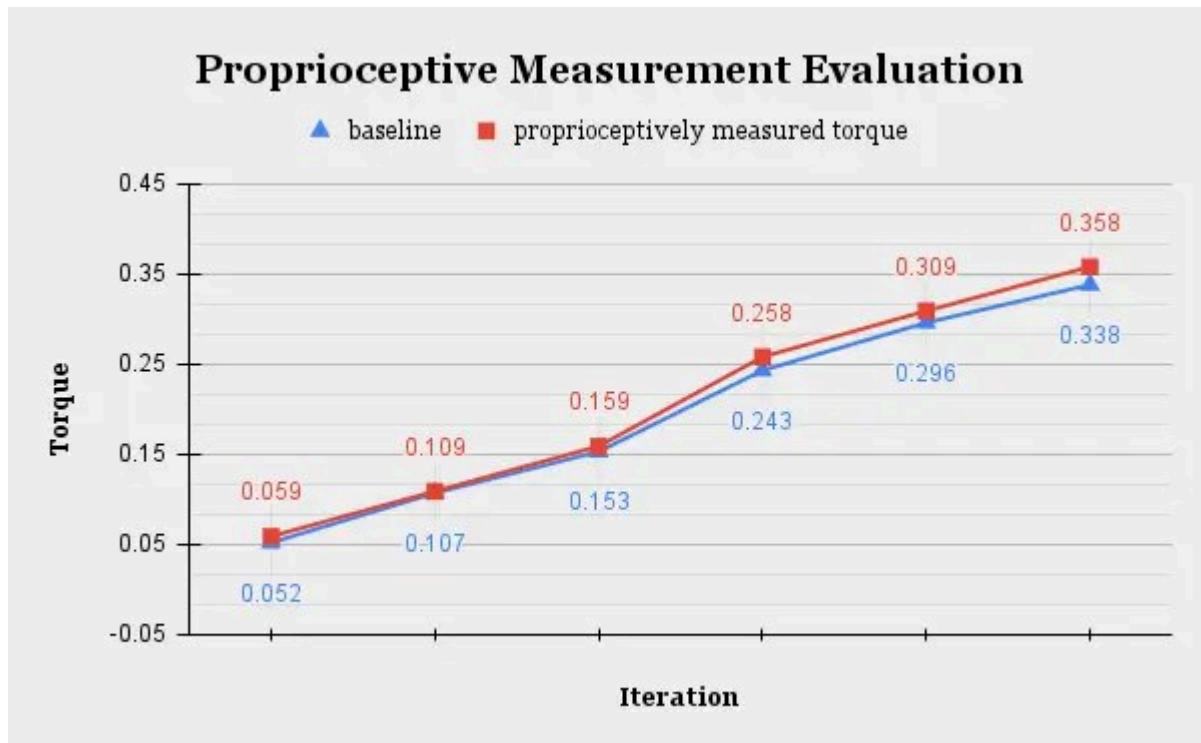
Proprioceptive Sensing Capacity

In the Clark/Park Transform which is the one of the process of FOC Algorithm, Three phase current converted to two current state; d-axis and q-axis current. q-axis current is proportional to torque that the motor produce. Thus, external torque(perturbance) can be estimated by measuring q-axis current of the motor.

Experiment

The experiment is conducted by 'Gartt ML5210 BLDC Drone Motor + DRV8302 + AS5600 Encoder + ESP-Wroom-32' and scale is used for baseline to validate the accuracy of the proprioceptive actuator.

Evaluation



Evaluation of Proprioceptive Sensing Capacity

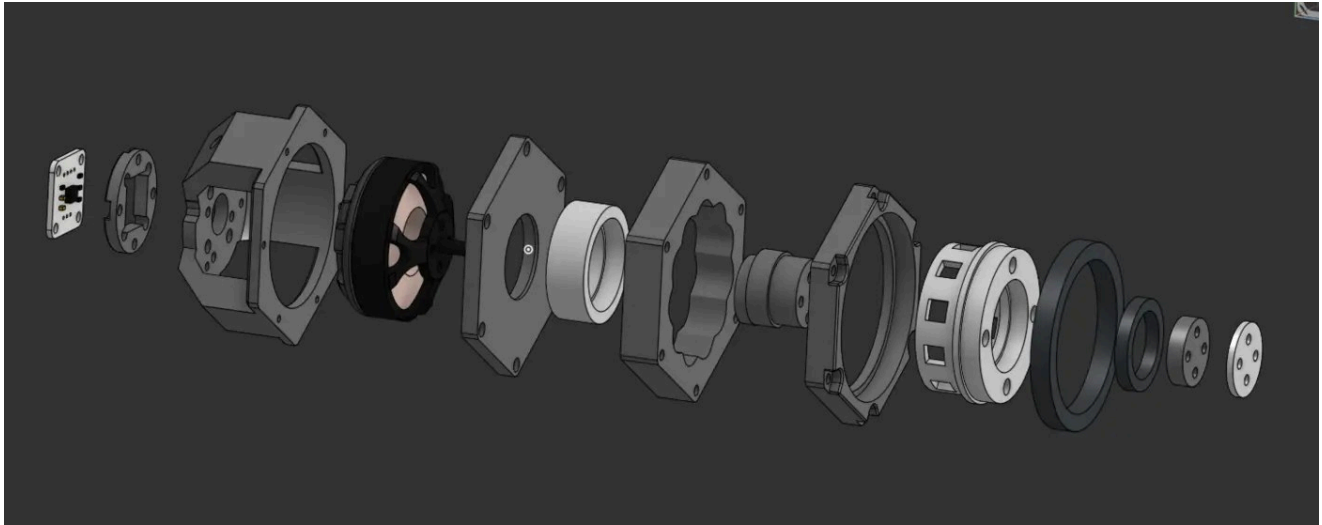
Approximately, the error is getting bigger as it produce higher torque.

CAN BUS Communication

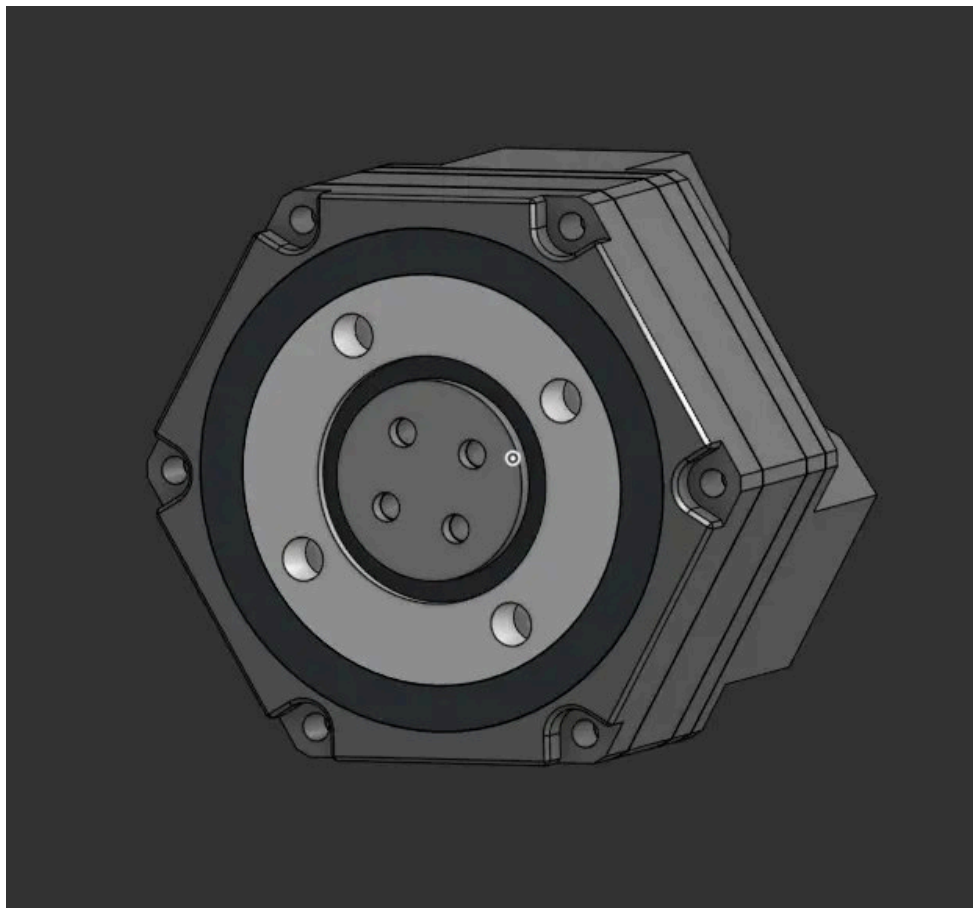
I used TJA1051T to enable CAN BUS Communication which enables communicate between other controllers(MCU). This is essential for using multiple actuators system like locomotive & manipulator systems.

BOM & resources for 3 Types of Actuators (Basic, Lite and Enhanced)

1. QDD Actuator Basic



Actuator image 2



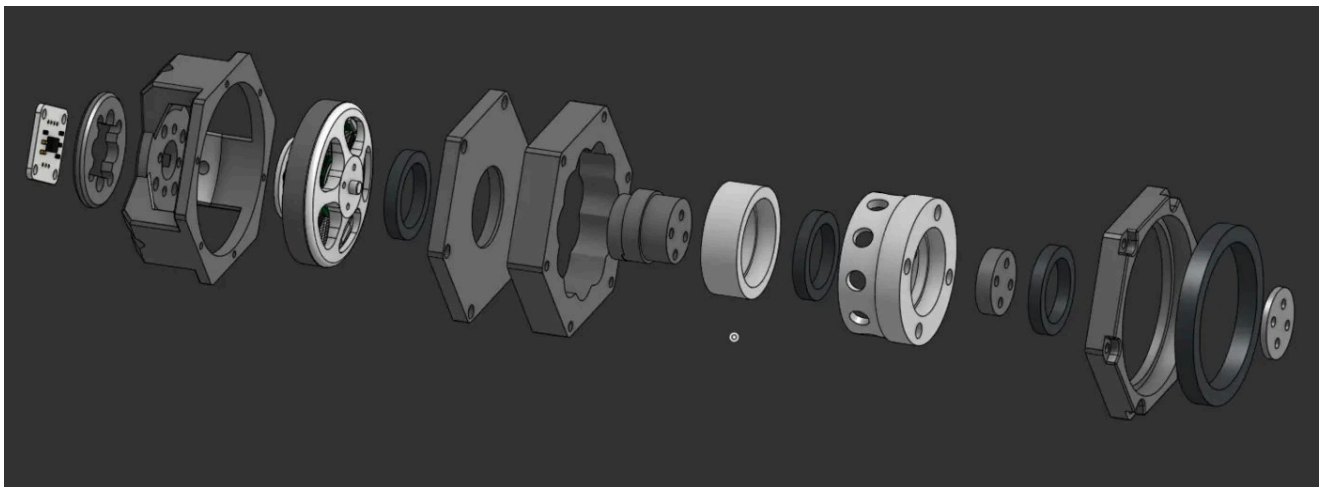
Actuator image 2

BOM of QDD Actuator Basic

Category	Component	Specification	Qty	Link
Motor & Drive	BLDC Motor	Sunnysky x4108s-17	1	View
Motor Driver	DRV8302	1	View	
Electronics	Microcontroller	ESP-WROOM-32	1	View
Magnetic Encoder	AS5600	1	View	
CAN Transceiver	TJA1051T	1	View	

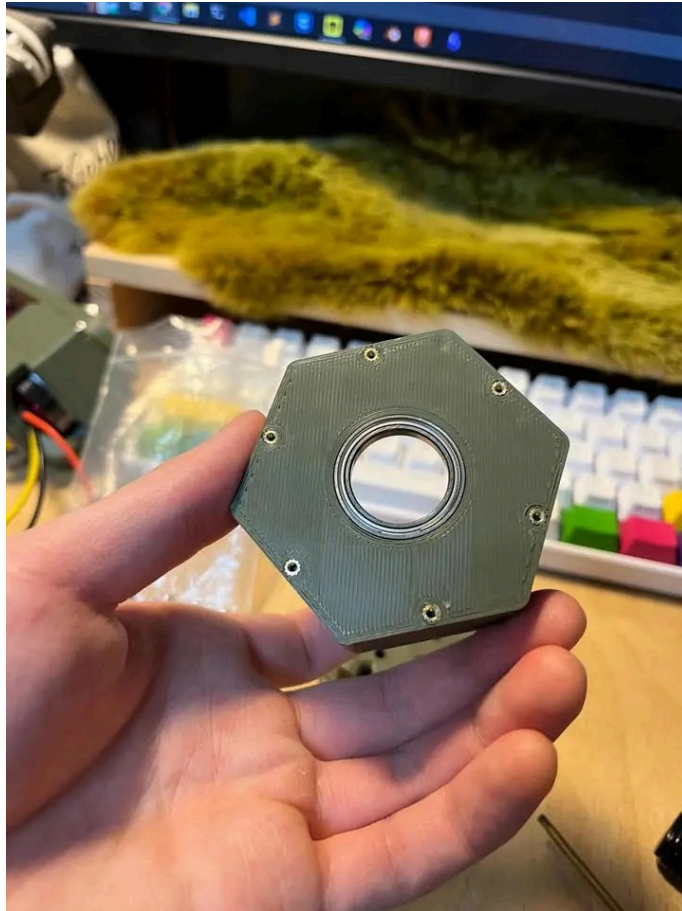
Mechanical	Gearbox	3D Printed Parts	1 set	-
Ball Elements	M6 x 6mm	10	View	
Bearings	20 x 27 x 4 mm	3	View	
Bearings	45 x 55 x 6 mm	1	View	
Fasteners	Bolts	M3 x 30 mm	4	View
Bolts	M3 x 20 mm	6	View	
Bolts	M2 x 10 mm	6	View	
Insert Nuts	M2 x 3 mm	6	View	
Insert Nuts	M3 x 3 mm	6	View	
Insert Nuts	M3 x 6 mm	6	View	

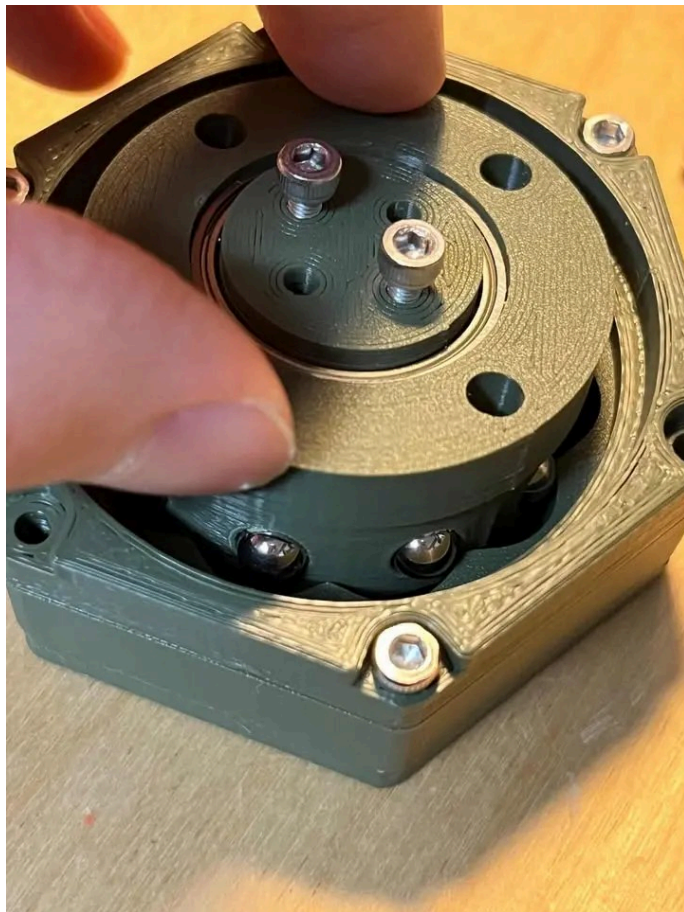
2. Cycloidal QDD Actuator Lite

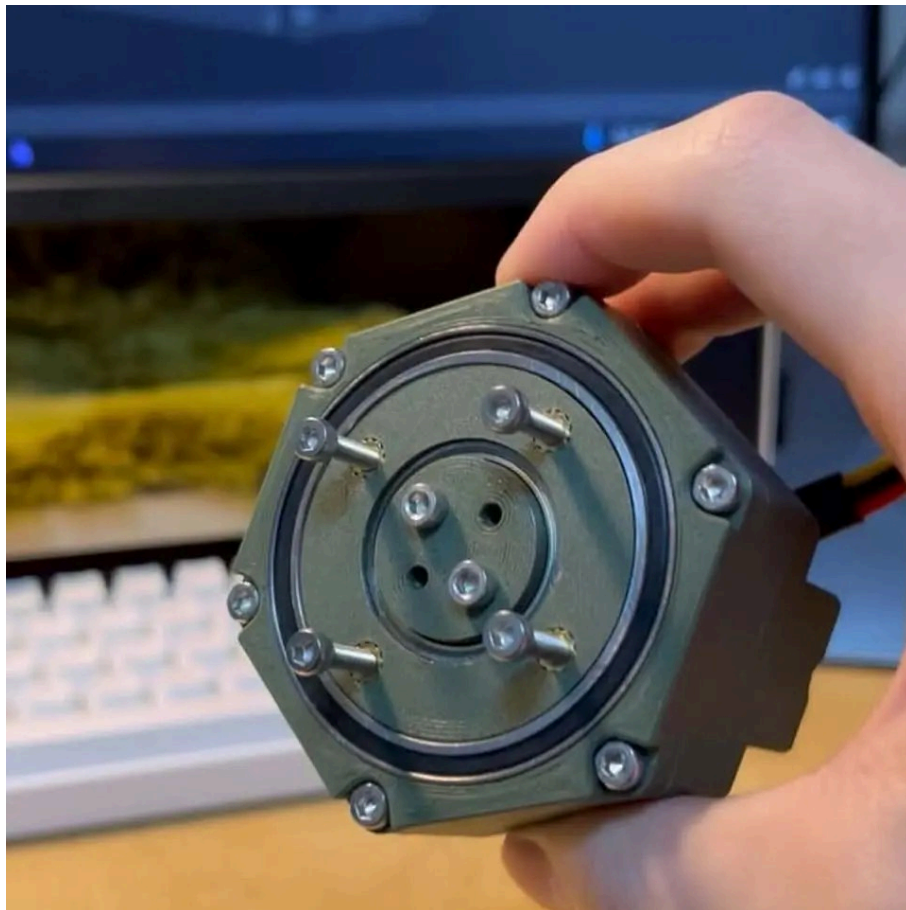
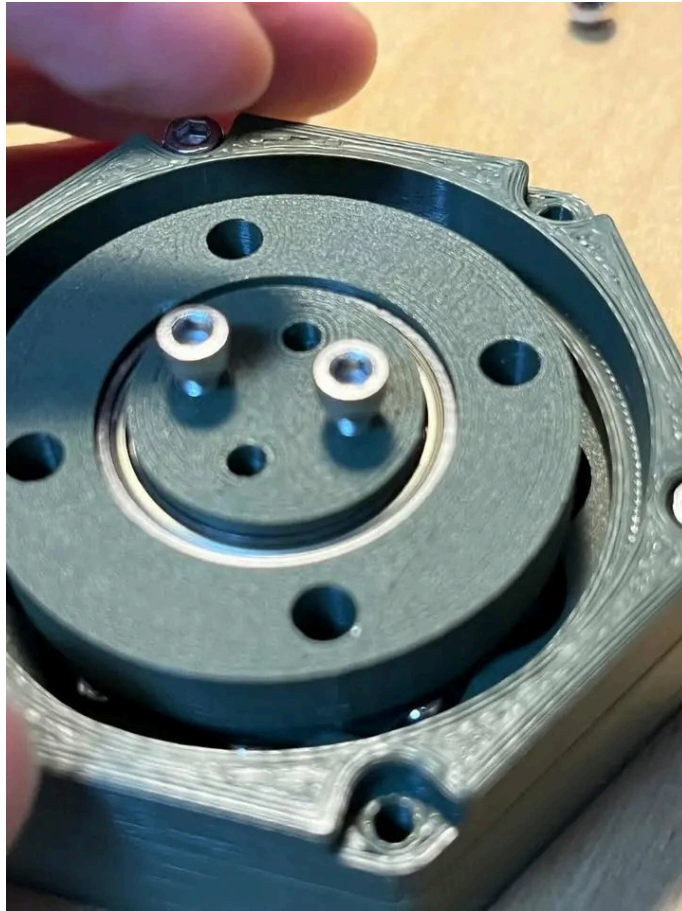


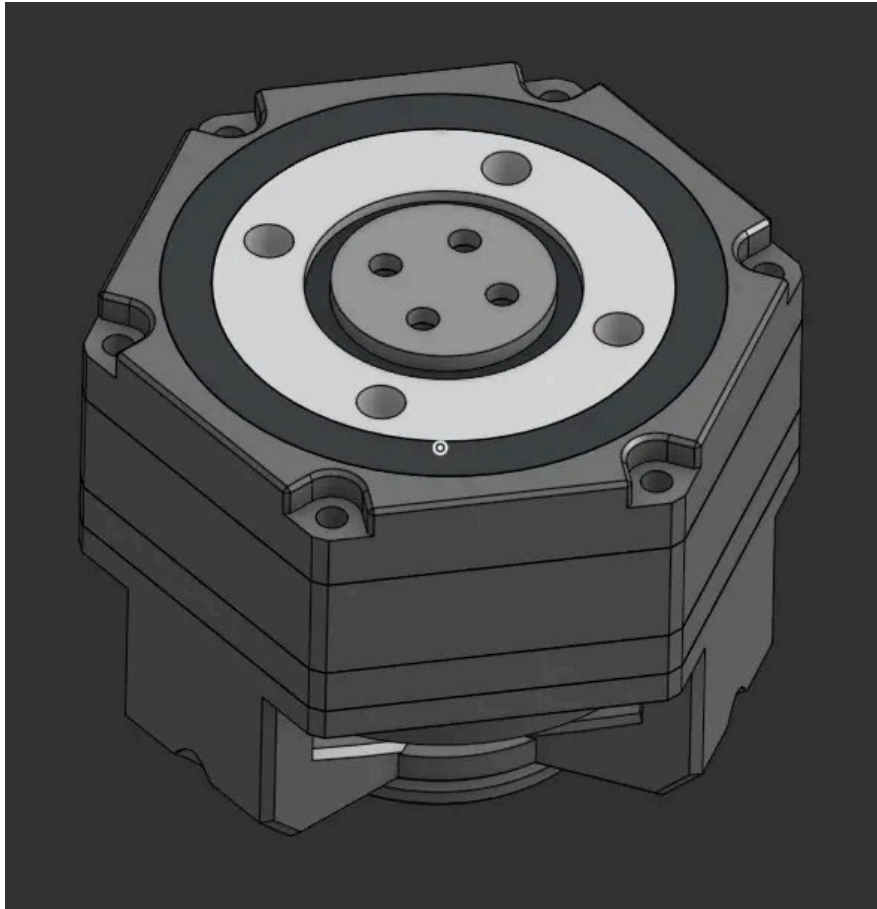
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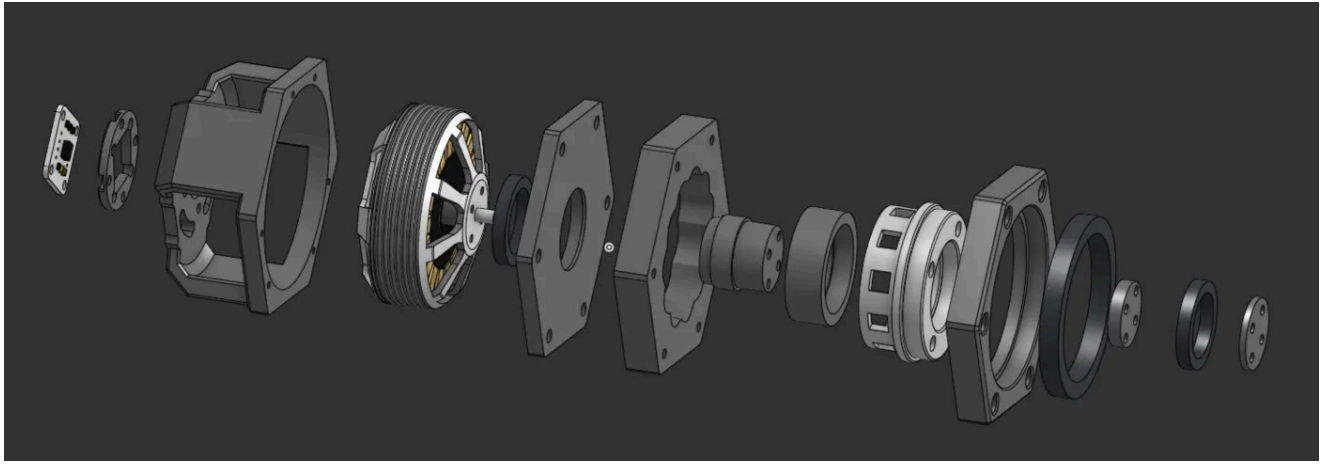


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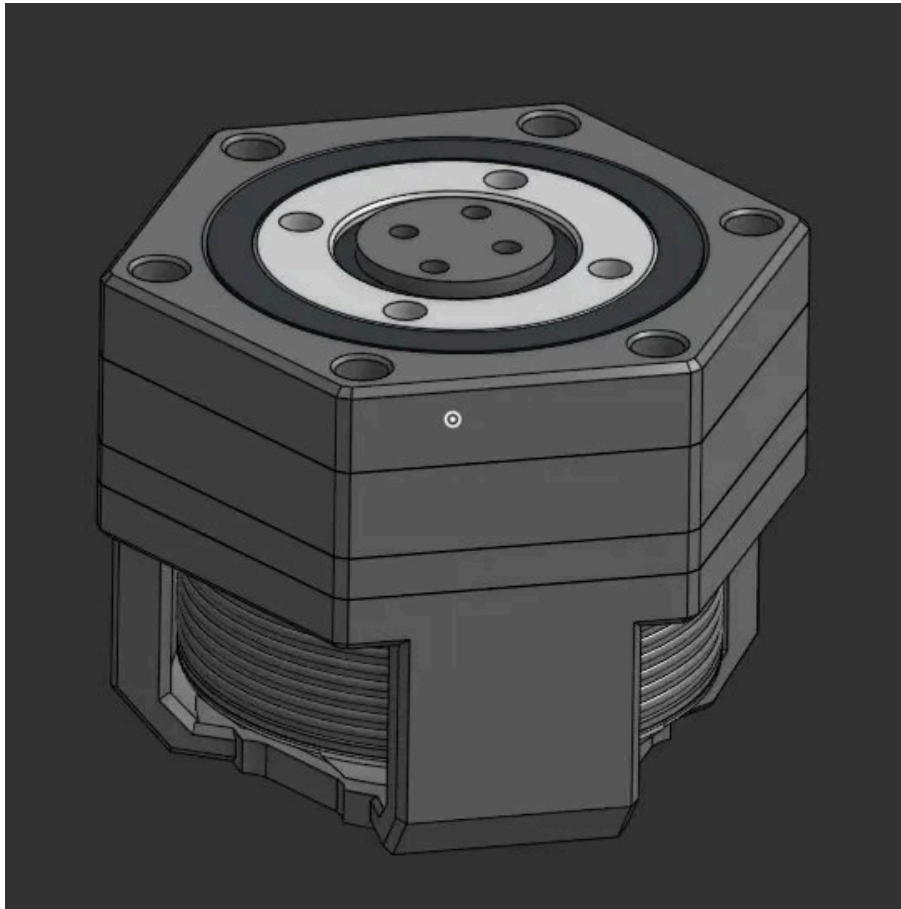
BOM of QDD Actuator Lite

Category	Component	Specification	Qty	Link
Motor & Drive	BLDC Motor	5010 BLDC Motor	1	View
Motor Driver	SimpleFOC Shield v2	1	View	
Electronics	Microcontroller	Arduino Uno	1	-
Magnetic Encoder	AS5600	1	View	
CAN Bus	None	-	-	
Mechanical	Gearbox	3D Printed Gearbox	1 set	-

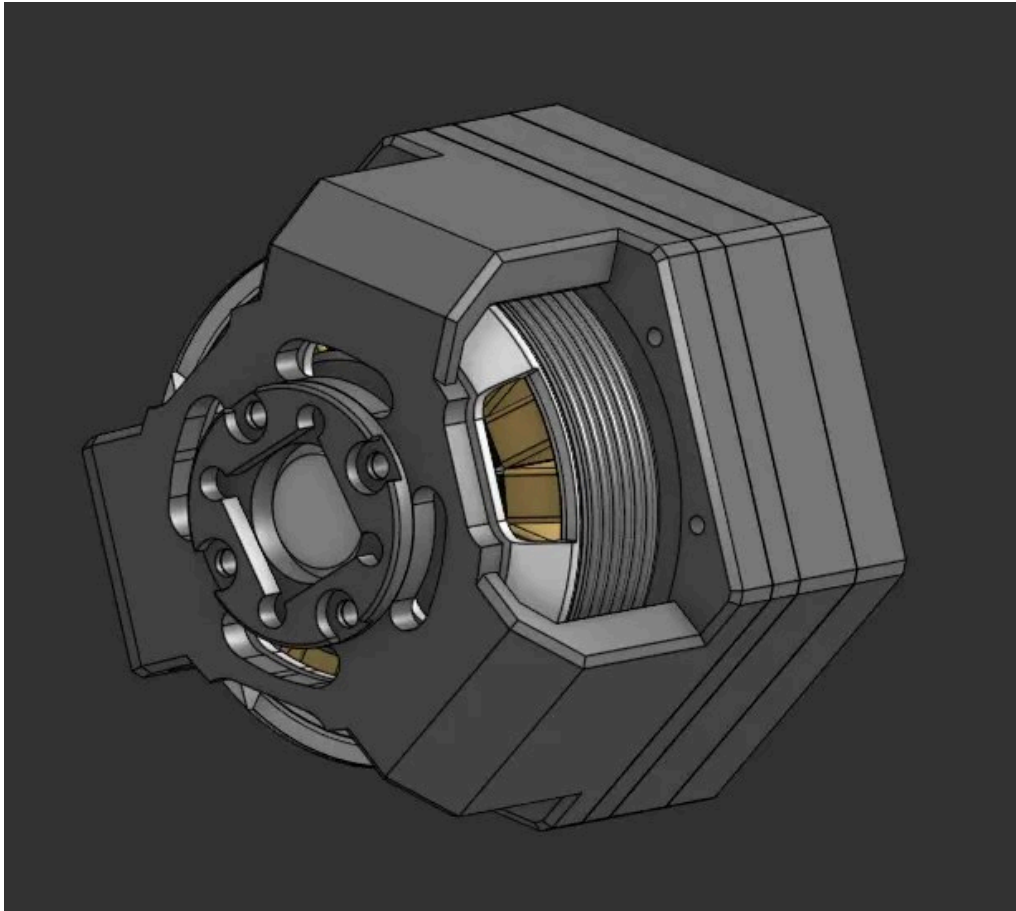
3. QDD Actuator Enhanced



Actuator image 2



Actuator image 2



Actuator image 2

BOM of QDD Actuator Enhanced

Category	Component	Specification	Qty	Link
Motor & Drive	BLDC Motor	Gartt ML5210	1	View
Motor Driver	DRV8302	1	View	
Electronics	Microcontroller	ESP-WROOM-32	1	View
Magnetic Encoder	AS5600	1	View	
CAN Transceiver	TJA1051T	1	View	
Mechanical	Gearbox	3D Printed Parts	1 set	-
Ball Elements	M6 x 6mm	10	View	
Bearings	20 x 27 x 4 mm	3	View	
Bearings	45 x 55 x 6 mm	1	View	
Fasteners	Bolts	M3 x 30 mm	4	View
Bolts	M3 x 20 mm	6	View	
Bolts	M2 x 10 mm	6	View	
Insert Nuts	M2 x 3 mm	6	View	
Insert Nuts	M3 x 3 mm	6	View	
Insert Nuts	M3 x 6 mm	6	View	

you can see more specific code at simplefoc documents

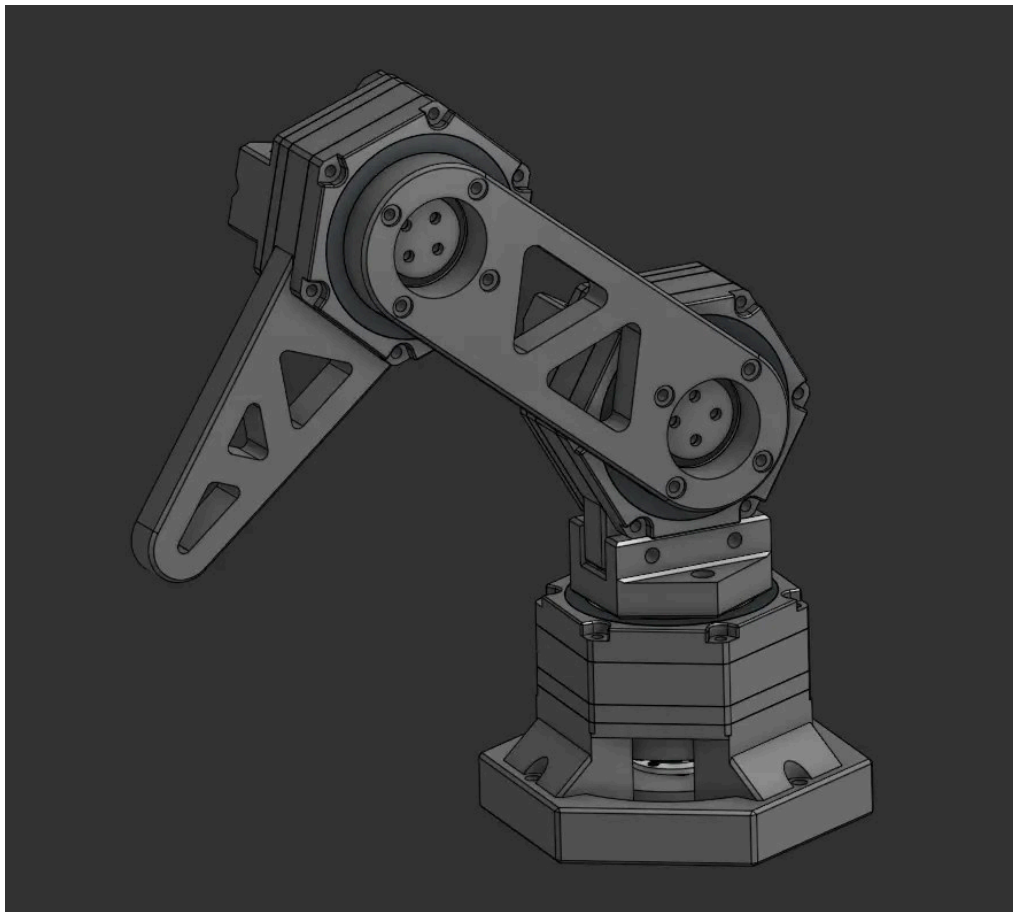
Applications of A Proprioceptive Actuator: 3DoF Robotic Arm & Humanoid Upper Body Robot through Impedance Control

출처: <https://jeongseojin.github.io/blog/2025/dof-robotic-arm-impedance-control/>

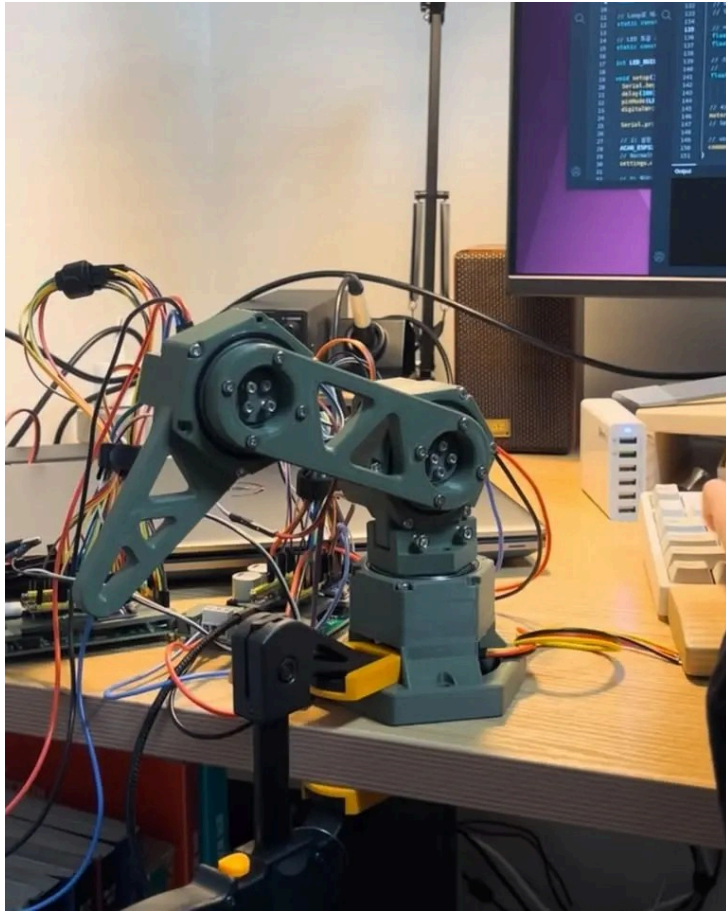
Background: Motivation and Evolution

Building upon my previous project, 'A Low-Cost Modular 3D-Printed Proprioceptive Actuator for HRI,' I developed a '3DoF Compliant Robotic Arm' and a 'Humanoid Upper Body Robot.' Both systems consist of fully 3D-printed parts and feature proprioceptive sensing capabilities, compliance, impedance control, and a QDD design.

3DoF Compliant Robotic Arm

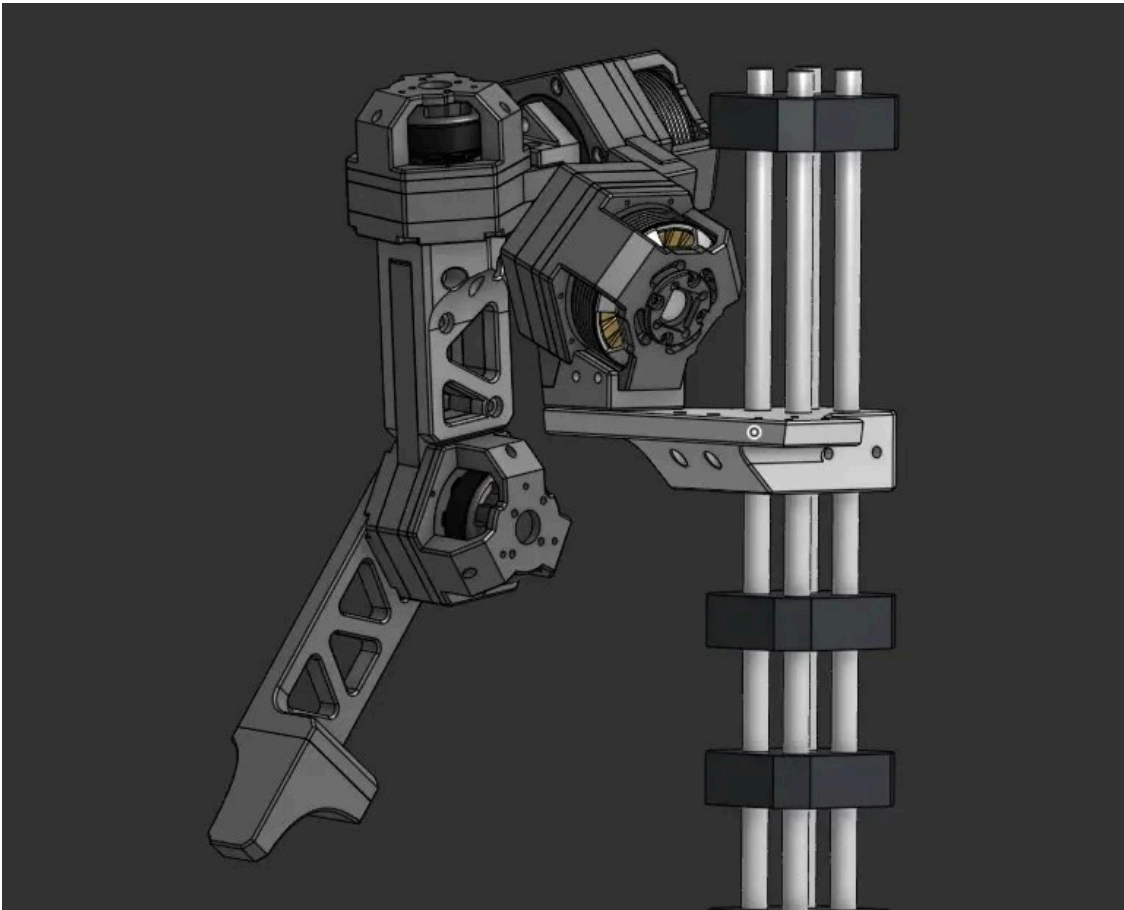


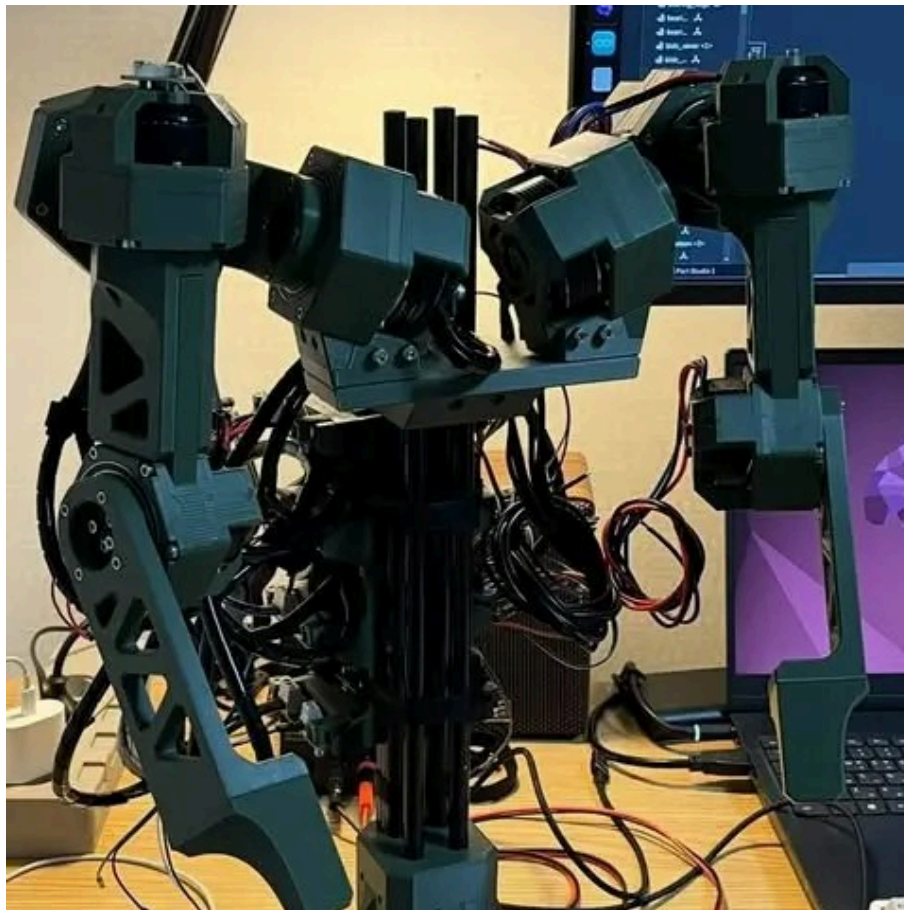
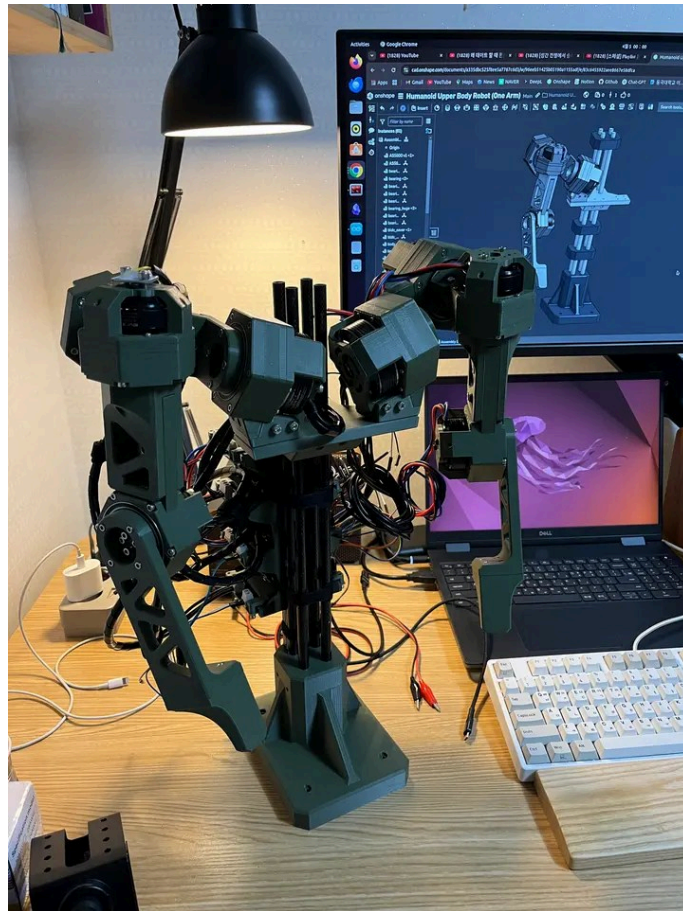




===== video =====

Humanoid Upper Body Robot





===== video =====

Conclusion

I built an integrated robotic system using my custom proprioceptive actuators and implemented impedance control, which allows the robot to react compliantly to external forces. Through this project, I demonstrated that my actuators are highly suitable for this robotic hardware platform. Although I have not yet implemented high-level controllers (such as Model Predictive Control or Reinforcement Learning), building this hardware from scratch was a significant milestone. It has accelerated my studies and provided a clear direction for my future projects.

Moving forward, I plan to use this platform to design vision-based controllers, MPC, and Reinforcement Learning models. This hardware will serve as a solid foundation for my extensive study of high-level control systems.

Paper Review: "Design Principles for Energy Efficient Legged Locomotion and Implementation on the MIT Cheetah Robot"

출처: <https://jeongseojin.github.io/blog/2025/review01/>

What is this paper regarding?

I reviewed this paper to gain insights into actuator design and overall energy management for legged locomotion to apply these principles to my robotic actuators. This paper represents three major energy-loss

1. Heat loss from actuator
2. Friction loss in transmission
3. Interaction loss between system and environment

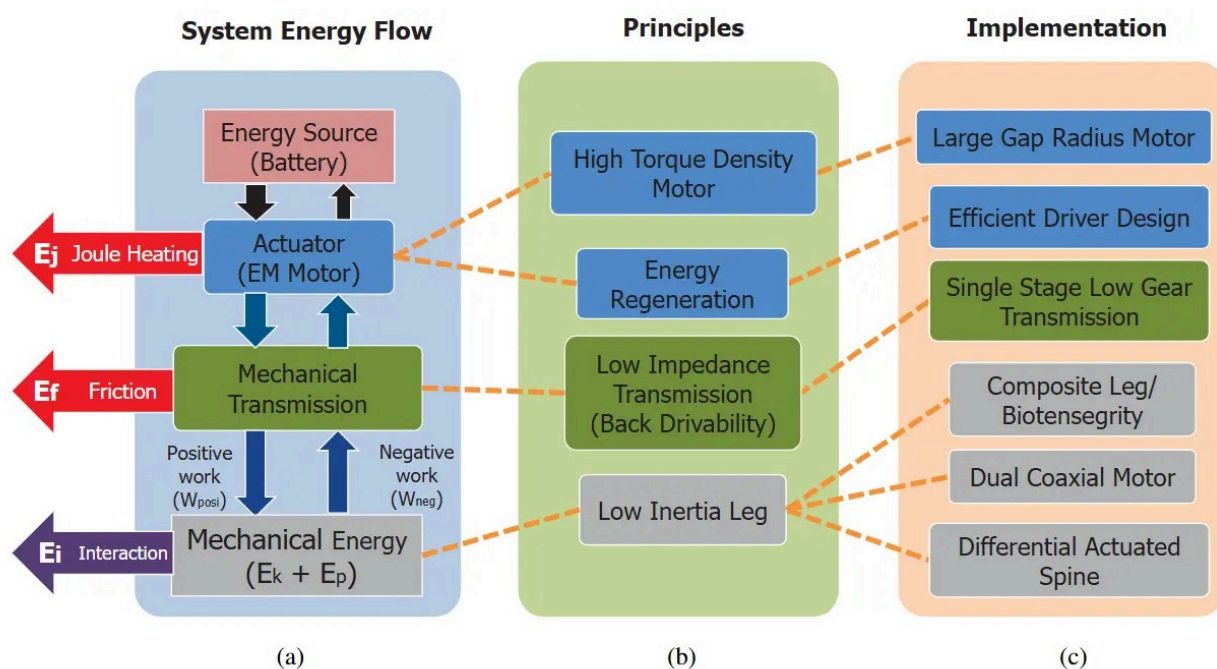


Fig. 2: (a) Energy flow diagram of the robot showing energy flows between the source and mechanical energy. Joule heating loss occurs at the motor, friction loss occurs in the mechanical transmission and interaction loss reduces the total mechanical energy. (b) Design principles to improve efficiency at the sources of energy loss. (c) Strategies for implementing the design principles for efficiency used on the MIT Cheetah Robot.

Energy flow diagram

$$E_{loss} = E_j + E_f + E_i$$

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+

I was particularly impressed by the intuitive analysis of the energy flow cycle in respect of actuator. Also, systematic methods to minimize these losses are well organized, such as employing

- High Torque Density Motor
- Energy Regenerative Electronic System
- Low Impedance Transmission(Back Drivability)
- Low Inertia Leg

It ultimately uses the Cost of Transport (

$\text{CoT} = \text{power} / \text{weight} \times \text{velocity}$

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$\times$

) as a key metric to evaluate the total energy efficiency of the locomotive system.

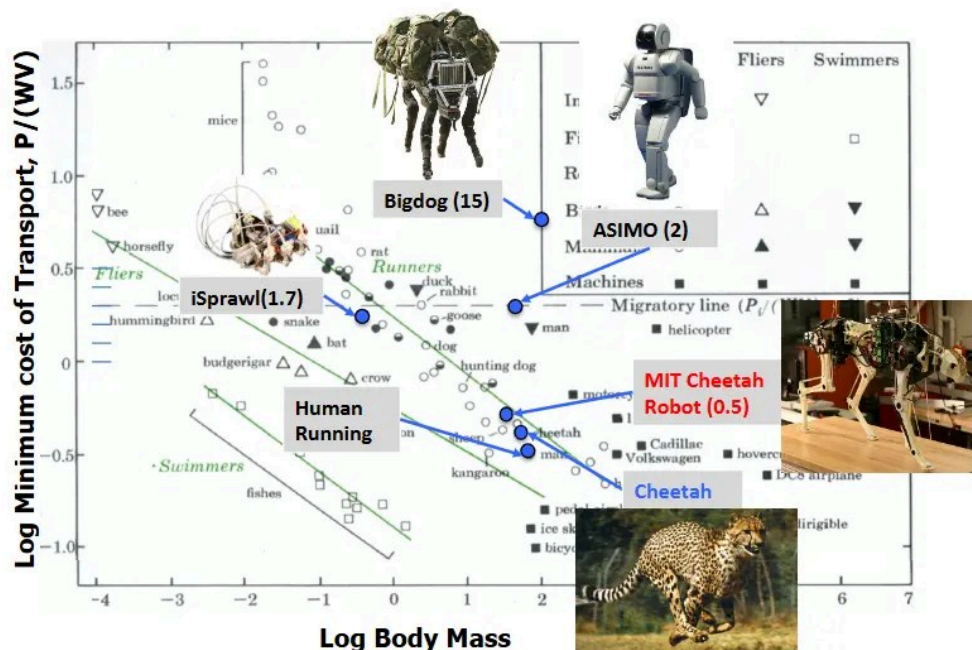


Fig. 1: Plot of total cost of transport (TCoT) vs body weight of animals and selected robots. The three classes of animal locomotion (running, flying, and swimming), occupy distinct regions of the plot. Used with permission from author of [2].

Actuator image 1

Key Contributions

- Energy Analysis : effectively reduced energy loss based on energy flow of the entire locomotive system.
- High Efficiency : MIT Cheetah achieved a cost of transport of 0.5 that rivals running animals and significantly lower than other running robots of its time.
- Proprioceptive Actuator Design : Demonstrated the effectiveness of low-gear ratio actuators(QDD) for dynamic interactions

Background of why I read this paper

In my previous project, Development of Cycloidal-QDD Actuator, I focused primarily on a mechanical implementation of my actuator to ensure the actuator worked. However, I realized the torque density and energy management system were not optimized. I lacked a comprehensive understanding of how individual actuator efficiency impacts the overall locomotive system.

This paper was perfect resource to bridge that gap. It clearly organizes the principle of energy-efficient actuator in dynamic locomotion. As I'm now focusing on building high-performance

actuators for successful dynamic control, understanding the principle applied to MIT Cheetah is crucial for my future work.

Critical Thinking & Takeaway

1. Energy Flow Diagram

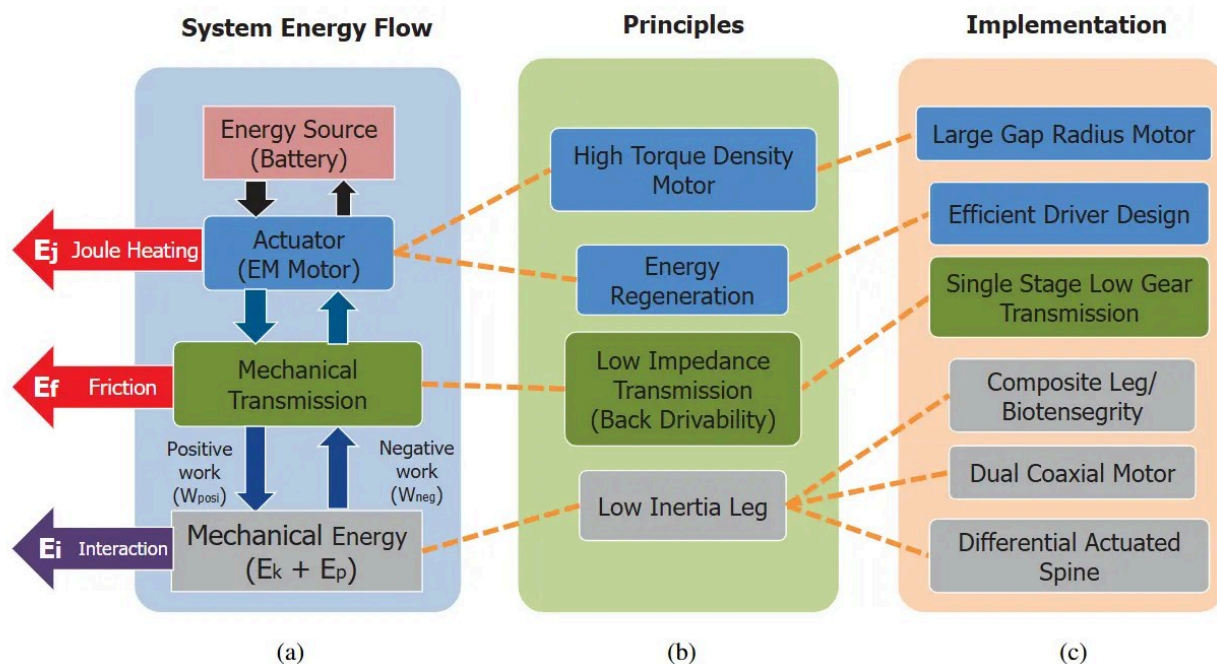


Fig. 2: (a) Energy flow diagram of the robot showing energy flows between the source and mechanical energy. Joule heating loss occurs at the motor, friction loss occurs in the mechanical transmission and interaction loss reduces the total mechanical energy. (b) Design principles to improve efficiency at the sources of energy loss. (c) Strategies for implementing the design principles for efficiency used on the MIT Cheetah Robot.

Energy flow diagram

Through above energy flow diagram in locomotion, I could analyze what exactly causes the energy loss in my actuator.

2. High Torque Density Motor(Large Gap Radius Motor) related to E_j

the electromagnetic(EM) motors with high-torque density minimize energy loss in actuators. Torque Density($\frac{\tau}{m}$) directly concerns the Joule heating of the EM motor by reducing the required electric current to provide torque.

Continuous torque of the motor represents how much torque the motor can generate at a constant heat dissipation. The continuous torque is directly related to motor constant, which represents torque per square root of Joule heating.

$$K_M = \tau \sqrt{I^2 R} \quad K_M = \tau \sqrt{I^2 R}$$

where:

- τ :torque
- I :current

- I:current
:
- R:resistance
:
\$\$

For example, if the torque density of the motor doubles without changing other factors, the Joule heating E_j can be reduced by 75%.

This means

KM

doubles, since torque doubles assuming mass remains constant

$$\tau = K M \times \sqrt{I^2 R} \quad \tau = K M \times I^2 R$$

Then to get a same torque, required current halved compared to the baseline. Therefore Joule Heating Loss(

$I^2 R$

$\frac{1}{2}$

) is a quarter of baseline which influences significant improved efficiency of the actuators

Parameter	Baseline	Improved	Logic
Torque Density	11	22	Doubled (Given condition)
Motor Constant (K_m)	K	2K2	Torque capacity doubles while mass stays constant
Required Current (I) (for same torque)	I_{req}	$\frac{1}{2}I_{req}$	Current is halved due to doubled K_m
Joule Heating Loss (E_j)	P_{loss}	$\frac{1}{4}P_{loss}$	Heating scales with current squared $(\frac{1}{2})^2 = \frac{1}{4}$ $(\frac{1}{2})^2 = \frac{1}{4}$
Reduction in Heat Loss	-	75% reduction	100 %- 25 %= 75 % 100 %- 25 %= 75 % saved

Consequently, increasing the torque density of the EM motor is highly desirable for efficiency reducing Joule Heating Loss(

E_j

)

Then, How can we get a high torque density EM motors? The torque density of the motor can be improved by increasing the gap radius: the radius of the gap between the motor stator windings and the permanent magnets on the rotor.

The torque density scales by approximately

$$\tau_m \propto r_{\text{gap}}$$

$\propto$

, the torque per inertia scales by

$$\tau/J \propto r_{\text{gap}}^{-1}$$

$\propto$

-1

, and the torque squared per electric power, also known as motor constant square, a measure of torque production efficiency, scales by $\tau^2/P = K^2 M \propto r_{\text{gap}}^3$. the results will favor the motor with the larger gap radius because it will have a smaller gear ratio and fewer gear-train stages; this results in less friction loss, higher torque density, and higher bandwidth.

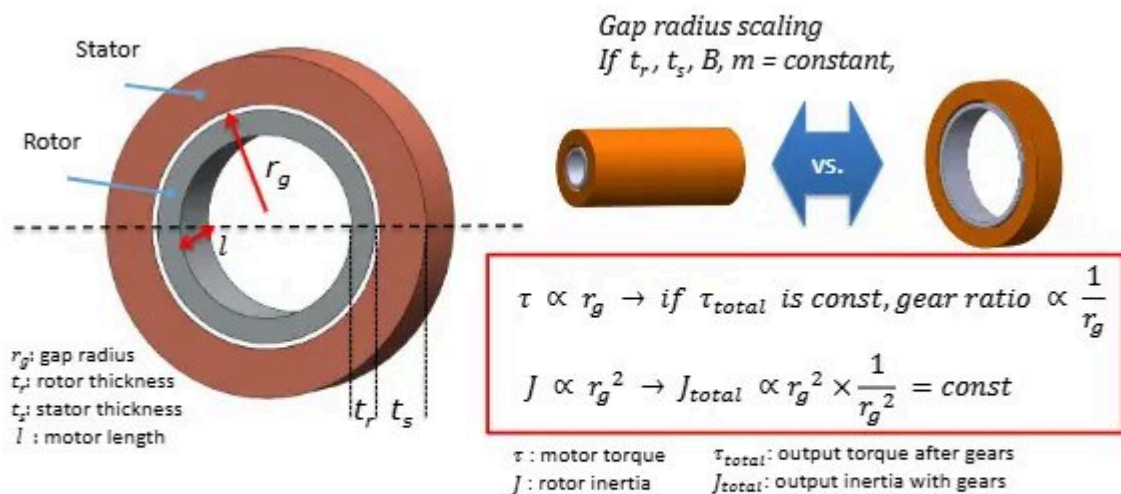


Fig. 5: Gap radius scaling of electromagnetic motor with a constant mass.

Figure5

3. Low Impedance Mechanical Transmission(low gear ratio transmission) related to Ef

Employing gears significantly reduces the torque demands on the motors while increasing torque density and reducing Joule Heating Loss(

Ej

). However, employing gears increase mechanical impedance like reflected inertia, reflected damping and gear friction while reducing back drivability, transmission transparency, efficiency. Further more, it may limit the control of the robot dynamics.

To minimize the lossess(

Ef

) associated with employment of gears, the low number of gear stages is inevitable especially in planetary gear which accumulate backlash stage after stage. And relatively low ratio reduces the contribution of reflected actuator dynamics on the mechanical impedance of the transmission. Therefore, low gear ratio is essential for efficient actuator design and the control for the dynamic robots

4. Low Inertia Leg

Low Inertia Leg is especially crucial for dynamic locomotion, such as higher stride frequency running increasing high bandwidth of the leg. Additionally Low Inertia leg mitigate impact loss(

E_i

) at touch down at every step. A large portion of the kinetic energy of the leg should be dissipated not only in the cyclic motion but also when the legs collide with the ground. therefore, employing low mass, inertia to the dynamic robots is critical for reducing energy loss caused by interaction between robot and environment.

Conclusion

In conclusion, understanding of the energy flow diagram is a crucial step for designing actuators. Based on these insights, Next version of my robotic actuator for dynamic robots will be designed minimizing the energy-losses mechanism employing high torque density EM motor, low impedance in transmissions and low inertia leg for future dynamic locomotive robots.

Reference

Citation: Seok, Sangok et al. "Design Principles for Energy-Efficient Legged Locomotion and Implementation on the MIT Cheetah Robot." IEEE/ASME Transactions on Mechatronics 20.3(2015): 1117–1129.

Publication: <http://dx.doi.org/10.1109/TMECH.2014.2339013>

NOTE: And this article is available as a open access article in DSpace@MIT!

Paper Review: "Demonstrating Berkeley Humanoid Lite: An Open-source, Accessible, and Customizable 3D-printed Humanoid Robot"

출처: <https://jeongseojin.github.io/blog/2026/review02/>

What is this paper regarding?

This paper demonstrates accessible 3d-printed low-cost humanoid robot promoting democratization in humanoid robotics. The core design is a modular 3d-printed actuator easily replaceable, preventing delays of an experiment from broken parts of robots. Also The authors conduct experiments to validate reliability of 3d-printed actuator. Furthermore, the experiments showcased zero-shot policy transfer from simulation to hardware, highlighting the platform's suitability for research validation.

Key Contributions

- Low Cost Accessible Modular 3d-printed actuator
- Actuator Evaluation of 3d-printed actuator
- Zero-shot policy transfer from simulation to hardware

Background of why I read this paper

Since I am planning to build a humanoid robot in the near future, the system must be reliable enough for successful Reinforcement Learning (RL) locomotion experiments, even if it is based on low-cost 3D-printed parts. Also, components should be quickly replaceable to accelerate the experimental cycle.

I'm now focusing on building reliable robotic actuator to employ reinforcement learning successfully. In my previous project, I did not conduct evaluations. Thus this paper is highly suitable for me, offering insights not only into actuator evaluation methods but also deploying reinforcement learning model on my robots.

Critical Thinking & Takeaway

1. Accessible Low-Cost Actuator

The actuators are two different sizes(6512, 5010). As shown in the figures below, It's relatively low cost compared to other metallic actuators. For example, 6512 actuator uses M6C12 150KV BLDC drone motor from MAD Components, AS5600 encoder and B-G431B-ESC1 as a motor driver. The reducer, and housing parts are all 3d printed with PLA. The cycloidal gears were selected as reducer due to its high robustness.

Motor Data			
Motor Model	MAD M6C12 EEE V2.0	Number of pole pairs	14
Stator	TAIWAN / Anticorrosive	Varnished wire Degree	180°C
Motor Size	D:72 × 27mm	Magnet Degree	150°C
Degree of Protection	Rain protection	Cable Length	60mm (extended Enameled wires)
Centrifugal Heat Dissipation	Independent	Rotor Balance	≤5 mg
Propeller Mounting Holes	D:12 M3×4, D:18 M3×4, D:22 M3×4	Motor Balance	≤10 mg
Shaft Diameter	IN: 6 mm	Motor Mounting Holes	D:25 M3×4, D:32 M4×4
Bearing	EZO 696ZZ *2	Disruptive test	500 V
Additional Accessories	M6 Prop Adapter *1, M6 Propeller Plate *1, Ø4-6 Adapter Ring *1, 3.5mm Bullet Connector*3, Heat Shrinkable Tube*3, M3*10mm *4 Motor Screws, M3*6mm *4 Prop Adapter Fixing Screws, M3*12mm *4 Propeller Screws, Sticker*1		
Specifications			
RPM/V	150KV	Nominal Voltage	12S lipo battery
No Load Current	0.67A / 20V	Internal resistance	90mΩ
Motor Weight	257 g	Product Boxed Weight	423g (110 x 110 x 55 mm)
Maximum Current	36.18 A	Maximum Power	1725.3W
Maximum thrust	9.4 kg	Maximum Torque	2.64 Nm
Recommended ESC	MAD AMPX 80A(5-14S)	Recommended Propellers	21x6.3, 22x6.6, 22.1x7.4, 24x7.2
UAV take-off weight	12S-22"/ 14kg--Quadcopter 21kg--Hexacopter 28kg--Octocopter	Single rotor take-off weight	2.0kg ~ 4.0kg

Actuator image 2

As you can see in the motor specification figure,

- Motor Size : D:72 x 27mm
- Number of the poles : 14
- RPM/V : 150KV
- Maximum Current : 36.18A
- Maximum Torque : 2.64Nm
- Nominal Battery : 12S lipo(

≈ ≈

48V)

2. Actuator Evaluation - Efficiency

A reliable actuator is fundamental to the robot's overall performance. This paper conduct a set of experiments under conditions identical to those on the robot, including 24V power supply, indential position PD gains, and matching position torque and current bandwidth configurations.

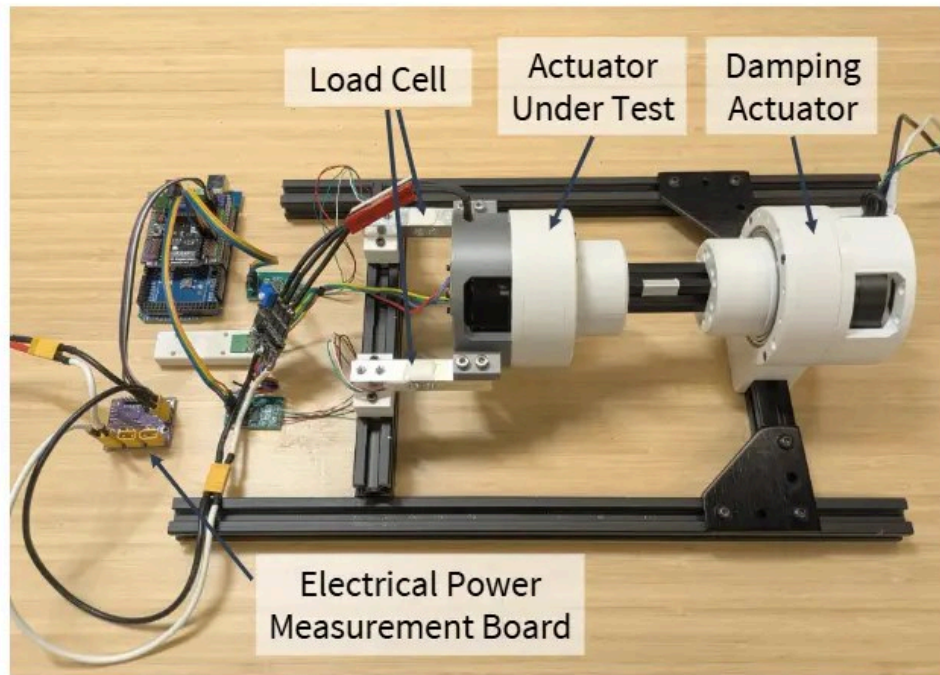


Fig. 7. Power efficiency test setup. Two load cells are used to directly measure the torque of the actuator under test, while another actuator acts as a damper, absorbing the mechanical energy produced by the actuator under test.

Actuator image 2

In the experiment, Actuator Under Test performs Torque(Torque Control) and Damping Actuator performs constant speed(speed control). Two load cells measure torque of the primary actuator. An electrical-power measurement board logged the supply voltage and current, from which electrical input power was calculated.

$$P M = \tau m \times \omega m$$

where:

- $P M$
: Mechanical Output Power
- τm
: Measured Torque
- ωm
: Measured Rotational Speed

Through these testset, Mechanical Output Power can be calculated which is power of the actuator.

$$M e f = P M \tau c \times v c$$

where:

- $M e f$
: Mechanical Efficiency
- τc
: Commanded Torque

- v_c
: Commanded Velocity

Then Mechanical Efficiency can be calculated either which represents the efficiency of 3d-printed reducer.

$$\eta_{eff} = \frac{P_{measured}}{P_{input}}$$

where:

- η_{eff}
: Total Efficiency
- $P_{measured}$
: Measured Mechanical Power
- P_{input}
: Input Power

Also, Total efficiency reflects the overall actuator efficiency, which includes motor copper losses, driver electrical losses, and mechanical losses.

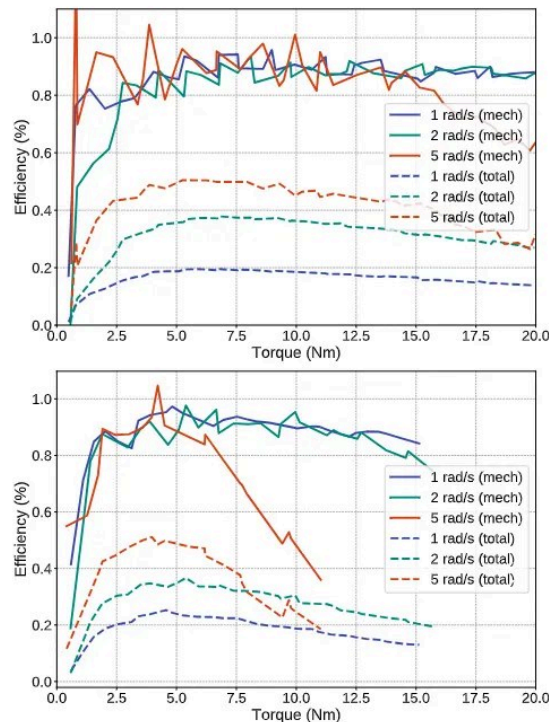


Fig. 8. Efficiency of the 6512 actuator (top), and the 5010 actuator (bottom). Mechanical efficiency is calculated as the ratio between the measured mechanical power to the product of the commanded torque and velocity, whereas total efficiency is defined as the ratio of the measured mechanical power to the electrical input power. The results indicate that, under most operating conditions, the gearbox maintains high efficiency.

Actuator image 2

The experiment conducted the actuator performance across three different speeds. Each torque and speed command was maintained for one second. As shown in Figure 8, the gearbox exhibits a mechanical efficiency of approximately 90% across most operating conditions. However, at high torque and velocity, efficiency decreases due to heat generation.

This dynamometer method is suitable for rigorous evaluation separating mechanical efficiency and total efficiency. we can easily analyze contributions of mechanical losses and electrical losses. This is

an effective method to evaluate custom actuators and reducers.

3. Actuator Evaluation - Transmission stiffness

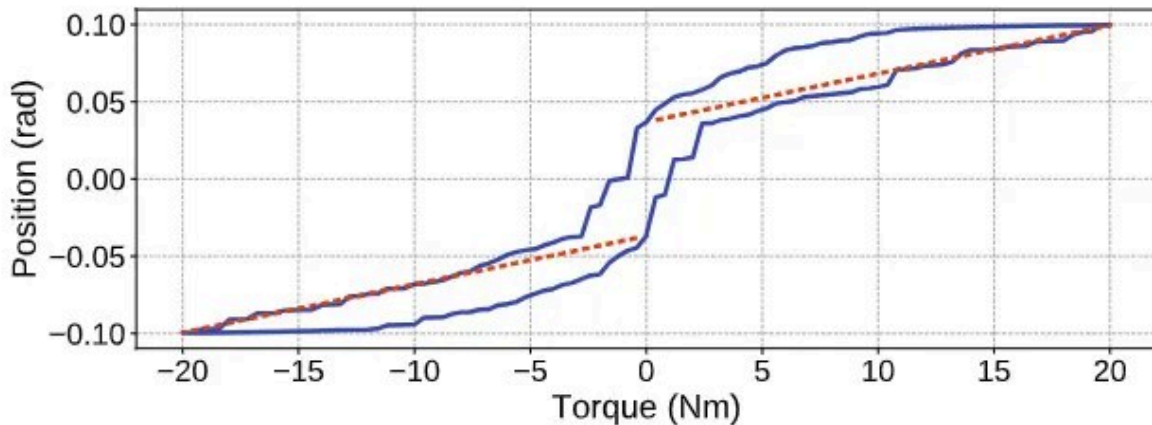


Fig. 9. Transmission stiffness measurement. The output of a 6512 actuator was fixed while the motor's command torque was gradually increased. The rotor position was recorded, yielding a mean stiffness of 319 N m.

Actuator image 2

The actuator's transmission stiffness is measured by rigidly fixing the output shaft relative to the actuator housing and measuring motor displacement under a range of static torques.

The torque command was gradually ramped from 0 Nm to 20 Nm and back in both directions. A linear fit was then applied to the data collected from 4 Nm to 10 Nm, and the inverse of the slope yielded a stiffness of approximately 319.49 Nm/rad (Figure 9). The transmission stiffness can be enhanced by opting alternative materials(e.g PA-CF).

4. Actuator - Durability

Durability is a primary consideration of 3d-printed actuator. The experiment was conducted for 60-hours lifting a pendulum(0.5kg, 0.5m) through range of -45 degrees to 90 degrees at a frequency of 0.5Hz. Consequently, efficiency and backlash remained within acceptable limits throughout the 60-hours test. The durability is remarkable especially considering cycloidal gear thickness is only 4mm. This finding gives me confidence in my own project, as my actuator design shares similar gear dimensions.

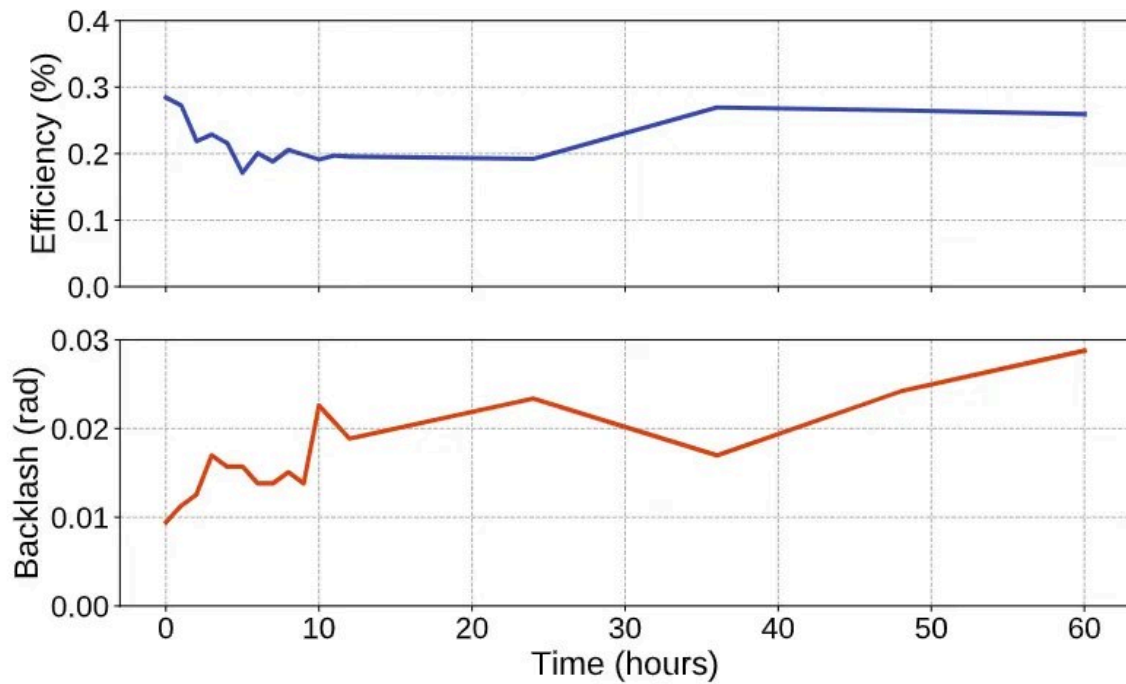


Fig. 10. Average total efficiency and backlash measured during the 60-hour durability test of the 6512 actuator. The efficiency remains stable throughout the test, while backlash progressively increases due to wear.

Actuator image 2

5. Legged Locomotion

This paper achieve a direct sim-to-real transfer of a Isaac Gym-trained policy to the physical robot without relying on additional state estimation methods. This is crucial for locomotive robot using 3d printed actuator which causes backlash. The authors formulate the locomotion task as a Partially Observed Markov Decision Process(POMDP) and use a standard Proximal Policy Optimization(PPO) algorithm to learn a control policy.

proprioceptive observations from the robot hardware:

- the base angular velocity
- the projected gravity vector
- joint positions and velocites

Additional inputs include the commanded linear velocity provided by the user and the previous time-step action. Notably, the experiement utilized only 30% of the actuator's torque limit, suggesting that the actuators are operating well within their capacity.

6. Additional Insights

Based on the specifications of the 6512 motor (2.64 Nm) and a gear ratio of 15:1 with an estimated efficiency of 90%, the theoretical maximum torque is approximately 35.64 Nm.

However, in this experimental setup, I estimate the practical maximum torque to be restricted to around 20–24 Nm. While the input voltage is 24V (compared to the motor's rated 48V), the primary limiting factor is likely a software-imposed current limit designed to protect the 3D-printed plastic gears from structural failure, rather than the motor's raw capability.

Conclusion

This paper presents that the low-cost 3d-printed actuator with zero-shot transfer policy, evaluating actuator's reliability and demonstrating the locomotion of the robot. Remarkably, the overall building/evaluation process of humanoid robot was impressive including actuator designs/evaluations and Reinforcement Learning deployment. Furthermore, 'Berkeley Humanoid Lite Docs' provide detailed documentation and instructions. Building the full humanoid robot with 3d-printed actuator, deploying the RL model on the robot, highly inspired me and boosted my confidence

Reference

Citation: <https://doi.org/10.48550/arXiv.2504.17249>

NOTE: This paper is available as an open-access paper

Berkeley Humanoid Lite Docs : <https://berkeley-humanoid-lite.gitbook.io/docs>

Image Sources

Controller image

Motor and Motor specification image

Encoder image